Cross sections and branching ratios

Table 1: Cross sections, branching ratios, uncertainties, and references.

Sample	Type	Value	Uncertainty	Reference	Comments
one_pb	σ [pb]	1	—	—	—
1pb	σ [pb]	1	—	—	—
BR_H_bb	BR	0.576	—	https://gitlab.cern.ch/hh/recommendations/-/blob/d7ad328c2ca55c75466b0a480af19823b7fa305b/CrossSections.md	—
BR_H_WW	BR	0.2203	—	https://gitlab.cern.ch/hh/recommendations/-/blob/d7ad328c2ca55c75466b0a480af19823b7fa305b/CrossSections.md	—
BR_H_ZZ	BR	0.02716	—	https://gitlab.cern.ch/hh/recommendations/-/blob/d7ad328c2ca55c75466b0a480af19823b7fa305b/CrossSections.md	—
BR_H_tautau	BR	0.06208	—	https://gitlab.cern.ch/hh/recommendations/-/blob/d7ad328c2ca55c75466b0a480af19823b7fa305b/CrossSections.md	—
BR_W_lnu	BR	0.1086	—	https://pdg.lbl.gov/2019/listings/rpp2019-list-w-boson.pdf	—
BR_W_qq	BR	0.8914 $= 1 - BR\_W\_lnu$	—	https://pdg.lbl.gov/2019/listings/rpp2019-list-w-boson.pdf	—
BR_Z_ll	BR	0.033658	—	https://pdg.lbl.gov/2018/listings/rpp2018-list-z-boson.pdf	—
BR_Z_nunu	BR	0.2	—	https://pdg.lbl.gov/2018/listings/rpp2018-list-z-boson.pdf	—
DY_NNLO_QCD_NLO_EW	σ [pb]	2094.2	$+0.7\%$ -1.1%	https://twiki.cern.ch/twiki/bin/viewauth/CMS/MATRIXCrossSectionsat13p6TeV	—
DYto2E_MLL_10to50_powheg	σ [pb]	6744	± 1.132	XSDBDYto2E_MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:40	—
DYto2E_MLL_50to120_powheg	σ [pb]	2219	± 0.2327	XSDBDYto2E_MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:19:01	—
DYto2E_MLL_120to200_powheg	σ [pb]	21.65	± 0.003184	XSDBDYto2E_MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:42	—
DYto2E_MLL_200to400_powheg	σ [pb]	3.058	± 0.000465	XSDBDYto2E_MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:48	—
DYto2E_MLL_400to800_powheg	σ [pb]	0.2691	$\pm 4.215e-05$	XSDBDYto2E_MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:57	—
DYto2E_MLL_800to1500_powheg	σ [pb]	0.01915	$\pm 3.085e-06$	XSDBDYto2E_MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:19:05	—
DYto2E_MLL_1500to2500_powheg	σ [pb]	0.001111	$\pm 1.787e-07$	XSDBDYto2E_MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:46	—
DYto2E_MLL_2500to4000_powheg	σ [pb]	$5.949e-05$	$\pm 9.162e-09$	XSDBDYto2E_MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:53	—
DYto2E_MLL_4000to6000_powheg	σ [pb]	$1.558e-06$	$\pm 2.078e-10$	XSDBDYto2E_MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:55	—
DYto2E_MLL_6000_powheg	σ [pb]	$3.519e-08$	$\pm 6.811e-12$	XSDBDYto2E_MLL-6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:19:03	—
DYto2Mu_MLL_50to130_powheg_minnlo	σ [pb]	2088	± 16.55	XSDBDYto2Mu_Bin-MLL-50to130_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:44:54	—
DYto2Mu_MLL_130to200_powheg_minnlo	σ [pb]	13.12	± 0.04125	XSDBDYto2Mu_Bin-MLL-130to200_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:44:54	—
DYto2Mu_MLL_200to400_powheg_minnlo	σ [pb]	3.058	± 0.000465	XSDBDYto2Mu_MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:11	—
DYto2Mu_MLL_400to600_powheg_minnlo	σ [pb]	0.2195	± 0.0006713	XSDBDYto2Mu_Bin-MLL-400to600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:46:01	—
DYto2Mu_MLL_600to800_powheg_minnlo	σ [pb]	0.04258	± 0.0001269	XSDBDYto2Mu_Bin-MLL-600to800_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:46:01	—
DYto2Mu_MLL_800to1000_powheg_minnlo	σ [pb]	0.01222	$\pm 3.782e-05$	XSDBDYto2Mu_Bin-MLL-800to1000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:46:01	—
DYto2Mu_MLL_1000to1500_powheg_minnlo	σ [pb]	0.006627	$\pm 2.065e-05$	XSDBDYto2Mu_Bin-MLL-1000to1500_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:44:56	—
DYto2Mu_MLL_1500to2000_powheg_minnlo	σ [pb]	0.0009183	$\pm 2.853e-06$	XSDBDYto2Mu_Bin-MLL-1500to2000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:44:44	—
DYto2Mu_MLL_2000to4000_powheg_minnlo	σ [pb]	0.0002396	$\pm 7.424e-07$	XSDBDYto2Mu_Bin-MLL-2000to4000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:45:35	—
DYto2Mu_MLL_4000to6000_powheg_minnlo	σ [pb]	$1.558e-06$	$\pm 2.078e-10$	XSDBDYto2Mu_MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:18	—
DYto2Mu_MLL_6000to13600_powheg_minnlo	σ [pb]	$3.519e-08$	$\pm 6.811e-12$	XSDBDYto2Mu_MLL-6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:25	—
DYto2L_M_10to50_amcatnloFXFX_allFlavors	σ [pb]	20950	± 183.5	XSDBDYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2023-03-2613:56:18	—

Continued on next page

Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
DYto2L_M_10to50_amcatnloFXFX_singleFlavor	σ [pb]	6983.3333 = $DYto2L\_M\_10to50\_amcatnloFXFX\_allFlavors/3$	±61.166667	XSDBDYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2023-03-2613:56:18	—
DYto2L_M_10to50_madgraphMLM	σ [pb]	17380	±26.57	XSDBDYto2L-4Jets_MLL-10to50_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:38	—
DYto2L_M_50_0J_amcatnloFXFX_allFlavors	σ [pb]	5378	±8.007	XSDBDYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:07	—
DYto2L_M_50_0J_amcatnloFXFX_singleFlavor	σ [pb]	1792.6667 = 5378.0/3	±2.669	XSDBDYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:07	divided by 3 to account for e, mu, tau
DYto2Tau_M_50_0J_Filtered_amcatnloFXFX_allFlavors	σ [pb]	847.035	±14.152	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_0J_Filtered_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22MiniAODv4-130X_mcRun3_2022_realistic_v5-v3/MINIAODSIM	—
DYto2Tau_M_50_0J_Filtered_amcatnloFXFX_singleFlavor	σ [pb]	282.345 = 847.035/3	±4.7173333	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_0J_Filtered_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22MiniAODv4-130X_mcRun3_2022_realistic_v5-v3/MINIAODSIM	divided by 3 to account for e, mu, tau
DYto2L_M_50_1J_amcatnloFXFX_allFlavors	σ [pb]	1017	±6.264	XSDBDYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:09	—
DYto2L_M_50_1J_amcatnloFXFX_singleFlavor	σ [pb]	339 = 1017.0/3	±2.088	XSDBDYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:09	divided by 3 to account for e, mu, tau
DYto2Tau_M_50_1J_Filtered_amcatnloFXFX_allFlavors	σ [pb]	210.112	±11.8	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_1J_Filtered_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22MiniAODv4-130X_mcRun3_2022_realistic_v5-v3/MINIAODSIM	—
DYto2Tau_M_50_1J_Filtered_amcatnloFXFX_singleFlavor	σ [pb]	70.037333 = 210.112/3	±3.9333333	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_1J_Filtered_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22MiniAODv4-130X_mcRun3_2022_realistic_v5-v3/MINIAODSIM	divided by 3 to account for e, mu, tau
DYto2L_M_50_2J_amcatnloFXFX_allFlavors	σ [pb]	385.5	±3.858	XSDBDYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:12	—
DYto2L_M_50_2J_amcatnloFXFX_singleFlavor	σ [pb]	128.5 = 385.5/3	±1.286	XSDBDYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:12	divided by 3 to account for e, mu, tau
DYto2Tau_M_50_2J_Filtered_amcatnloFXFX_allFlavors	σ [pb]	83.268	±8.875	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_2J_Filtered_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22MiniAODv4-130X_mcRun3_2022_realistic_v5-v4/MINIAODSIM	—
DYto2Tau_M_50_2J_Filtered_amcatnloFXFX_singleFlavor	σ [pb]	27.756 = 83.268/3	±2.9583333	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_2J_Filtered_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22MiniAODv4-130X_mcRun3_2022_realistic_v5-v4/MINIAODSIM	divided by 3 to account for e, mu, tau
DYto2L_M_50_1J_madgraphMLM_allFlavors	σ [pb]	973.1	±2.613	XSDBDYto2L-4Jets_MLL-50_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:40	—
DYto2L_M_50_1J_madgraphMLM_singleFlavor	σ [pb]	324.36667 = 973.1/3	±0.871	XSDBDYto2L-4Jets_MLL-50_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:40	—
DYto2L_M_50_2J_madgraphMLM_allFlavors	σ [pb]	312.4	±0.915	XSDBDYto2L-4Jets_MLL-50_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:42	—
DYto2L_M_50_2J_madgraphMLM_singleFlavor	σ [pb]	104.13333 = 312.4/3	±0.305	XSDBDYto2L-4Jets_MLL-50_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:42	—
DYto2L_M_50_3J_madgraphMLM_allFlavors	σ [pb]	93.93	±0.2858	XSDBDYto2L-4Jets_MLL-50_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:44	—
DYto2L_M_50_3J_madgraphMLM_singleFlavor	σ [pb]	31.31 = 93.93/3	±0.095266667	XSDBDYto2L-4Jets_MLL-50_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:44	—
DYto2L_M_50_4J_madgraphMLM_allFlavors	σ [pb]	45.43	±0.1393	XSDBDYto2L-4Jets_MLL-50_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:46	—
DYto2L_M_50_4J_madgraphMLM_singleFlavor	σ [pb]	15.143333 = 45.43/3	±0.046433333	XSDBDYto2L-4Jets_MLL-50_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:46	—
DYto2L_M_50_amcatnloFXFX_allFlavors	σ [pb]	6688	±83.99	XSDBDYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2023-03-2613:56:19	—
DYto2L_M_50_amcatnloFXFX_singleFlavor	σ [pb]	2229.3333 = 6688.0/3	±27.996667	XSDBDYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2023-03-2613:56:19	divided by 3 to account for e, mu, tau
DYto2L_M_50_madgraphMLM_allFlavors	σ [pb]	5467	±13.22	XSDBDYto2L-4Jets_MLL-50_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:49	—
DYto2L_M_50_madgraphMLM_singleFlavor	σ [pb]	1822.3333 = 5467.0/3	±4.4066667	XSDBDYto2L-4Jets_MLL-50_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:49	—
DYto2L_M_50_PTLL_40to100_1J_amcatnloFXFX_allFlavors	σ [pb]	475.3	±2.811	XSDBDYto2L-2Jets_MLL-50_PTLL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:27	—
DYto2L_M_50_PTLL_40to100_1J_amcatnloFXFX_singleFlavor	σ [pb]	158.43333 = 475.3/3	±0.937	XSDBDYto2L-2Jets_MLL-50_PTLL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:27	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_40to100_2J_amcatnloFXFX_allFlavors	σ [pb]	179.3	±1.827	XSDBDYto2L-2Jets_MLL-50_PTLL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:29	—
DYto2L_M_50_PTLL_40to100_2J_amcatnloFXFX_singleFlavor	σ [pb]	59.766667 = 179.3/3	±0.609	XSDBDYto2L-2Jets_MLL-50_PTLL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:29	divided by 3 to account for e, mu, tau

Continued on next page

Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
DYto2L_M_50_PTLL_100to200_1J_amcatnloFXFX_allFlavors	σ [pb]	45.42	± 0.2314	XSDBDYto2L-2Jets_MLL-50_PTLL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:15	—
DYto2L_M_50_PTLL_100to200_1J_amcatnloFXFX_singleFlavor	σ [pb]	15.14 $= 45.42/3$	± 0.077133333	XSDBDYto2L-2Jets_MLL-50_PTLL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:15	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_100to200_2J_amcatnloFXFX_allFlavors	σ [pb]	51.68	± 0.4864	XSDBDYto2L-2Jets_MLL-50_PTLL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:16	—
DYto2L_M_50_PTLL_100to200_2J_amcatnloFXFX_singleFlavor	σ [pb]	17.226667 $= 51.68/3$	± 0.16213333	XSDBDYto2L-2Jets_MLL-50_PTLL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:16	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_200to400_1J_amcatnloFXFX_allFlavors	σ [pb]	3.382	± 0.01443	XSDBDYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:18	—
DYto2L_M_50_PTLL_200to400_1J_amcatnloFXFX_singleFlavor	σ [pb]	1.1273333 $= 3.382/3$	± 0.00481	XSDBDYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:18	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_200to400_2J_amcatnloFXFX_allFlavors	σ [pb]	7.159	± 0.05581	XSDBDYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:19	—
DYto2L_M_50_PTLL_200to400_2J_amcatnloFXFX_singleFlavor	σ [pb]	2.3863333 $= 7.159/3$	± 0.018603333	XSDBDYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:19	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_400to600_1J_amcatnloFXFX_allFlavors	σ [pb]	0.1162	± 0.0004367	XSDBDYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:21	—
DYto2L_M_50_PTLL_400to600_1J_amcatnloFXFX_singleFlavor	σ [pb]	0.038733333 $= 0.1162/3$	± 0.00014556667	XSDBDYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:21	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_400to600_2J_amcatnloFXFX_allFlavors	σ [pb]	0.4157	± 0.002364	XSDBDYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:23	—
DYto2L_M_50_PTLL_400to600_2J_amcatnloFXFX_singleFlavor	σ [pb]	0.13856667 $= 0.4157/3$	± 0.000788	XSDBDYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:23	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_600_1J_amcatnloFXFX_allFlavors	σ [pb]	0.01392	$\pm 5.797e-05$	XSDBDYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:34	—
DYto2L_M_50_PTLL_600_1J_amcatnloFXFX_singleFlavor	σ [pb]	0.00464 $= 0.01392/3$	5.797e-05 / 3	XSDBDYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:34	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_600_2J_amcatnloFXFX_allFlavors	σ [pb]	0.07019	± 0.0003334	XSDBDYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:36	—
DYto2L_M_50_PTLL_600_2J_amcatnloFXFX_singleFlavor	σ [pb]	0.023396667 $= 0.07019/3$	± 0.00011113333	XSDBDYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:36	divided by 3 to account for e, mu, tau
DYto2Mu_MLL_105to160_amcatnloFXFX	σ [pb]	49.03	± 0.209	XSDBDYto2Mu-2Jets_MLL-105To160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-12-2007:59:27	—
DYto2Mu_MLL_105to160_amcatnloFXFX_VBFFilter	σ [pb]	3.105	± 0.006282	XSDBDYto2Mu-2Jets_Bin-MLL-105to160_Fil-VBF_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2026-05-0614:44:56	—
DYToMuMu-M105-180_13p6TeV_MiNNLO_pythia8	σ [pb]	48.6	± 0.01	internalfromprivateprod	—
DYto2Mu_MLL_10to50_powheg	σ [pb]	6744	± 1.132	XSDBDYto2Mu_MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:02	—
DYto2Mu_MLL_50to120_powheg	σ [pb]	2219	± 0.2327	XSDBDYto2Mu_MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:24	—
DYto2Mu_MLL_120to200_powheg	σ [pb]	21.65	± 0.003184	XSDBDYto2Mu_MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:04	—
DYto2Mu_MLL_200to400_powheg	σ [pb]	3.058	± 0.000465	XSDBDYto2Mu_MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:11	—
DYto2Mu_MLL_400to800_powheg	σ [pb]	0.2691	$\pm 4.215e-05$	XSDBDYto2Mu_MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:21	—
DYto2Mu_MLL_800to1500_powheg	σ [pb]	0.01915	$\pm 3.085e-06$	XSDBDYto2Mu_MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:27	—
DYto2Mu_MLL_1500to2500_powheg	σ [pb]	0.001111	$\pm 1.787e-07$	XSDBDYto2Mu_MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:10	—
DYto2Mu_MLL_2500to4000_powheg	σ [pb]	$5.949e-05$	$\pm 9.162e-09$	XSDBDYto2Mu_MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:16	—
DYto2Mu_MLL_4000to6000_powheg	σ [pb]	$1.558e-06$	$\pm 2.078e-10$	XSDBDYto2Mu_MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:18	—
DYto2Mu_MLL_6000_powheg	σ [pb]	$3.519e-08$	$\pm 6.811e-12$	XSDBDYto2Mu_MLL-6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:25	—
DYto2Tau_MLL_10to50_powheg	σ [pb]	6744	± 1.132	XSDBDYto2Tau_MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:29	—
DYto2Tau_MLL_50to120_powheg	σ [pb]	2219	± 0.2327	XSDBDYto2Tau_MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:45	—
DYto2Tau_MLL_120to200_powheg	σ [pb]	21.65	± 0.003184	XSDBDYto2Tau_MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:31	—
DYto2Tau_MLL_200to400_powheg	σ [pb]	3.058	± 0.000465	XSDBDYto2Tau_MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:36	—
DYto2Tau_MLL_400to800_powheg	σ [pb]	0.2691	$\pm 4.215e-05$	XSDBDYto2Tau_MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:43	—

Continued on next page

Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
DYto2Tau_MLL_800to1500_powheg	σ [pb]	0.01915	$\pm 3.085e-06$	XSDBDYto2Tau_MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:49	—
DYto2Tau_MLL_1500to2500_powheg	σ [pb]	0.001111	$\pm 1.787e-07$	XSDBDYto2Tau_MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:34	—
DYto2Tau_MLL_2500to4000_powheg	σ [pb]	$5.949e-05$	$\pm 9.162e-09$	XSDBDYto2Tau_MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:38	—
DYto2Tau_MLL_4000to6000_powheg	σ [pb]	$1.558e-06$	$\pm 2.078e-10$	XSDBDYto2Tau_MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:41	—
DYto2Tau_MLL_6000_powheg	σ [pb]	$3.519e-08$	$\pm 6.811e-12$	XSDBDYto2Tau_MLL-6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:47	—
DYto2TautoMuTauh_M_50_madgraphMLM	σ [pb]	53.28	± 0.5212	XSDBDYto2TautoMuTauh_M-50_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-03-2613:56:19	—
LLJJ_SM_M105-160_MJJ120_13p6TeV_dipole	σ [pb]	0.06	—	internalfromprivateprod	—
LLJJ_SM_M105-160_MJJ120_13p6TeV_dipole_vbffilter	σ [pb]	0.06	—	internalfromprivateprod	—
LLJJ_SM_M-50_MJJ120_13p6TeV_dipole	σ [pb]	0.88	—	internalfromprivateprod	—
LLJJ_SM_M-50_MJJ120_13p6TeV_herwig7	σ [pb]	0.88	—	internalfromprivateprod	—
GluGluHto2Mu_M120	σ [pb]	0.013595453 $= 56.11 \times (0.0002423)$	+4.6% -6.9% (theory) +-4% (TH Gaussian) +-3.2% (PDF+alpha s) +-1.9% (PDF) +-2.6% (alpha s) for XS ; +1.23% 1.23% (theory) +1.07% -1.09% (mQ) +0.69% -0.70% (alphas) for BR	https://arxiv.org/pdf/2402.09955forXSandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGluHto2Mu	σ [pb]	0.011365248 $= 52.23 \times (0.0002176)$	+4.6% -6.7% (theory) +-3.9 (TH Gaussian) +-3.2 (PDF+alpha s) +-1.9 (PDF) +-2.6 (alpha s) ; +1.23% -1.2% (theory) +0.97% -0.99% (mq) +0.59% -0.64% (alphas)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGluHto2Mu_M125	σ [pb]	0.011365248 $= 52.23 \times (0.0002176)$	+4.6% -6.7% (theory) +-3.9 (TH Gaussian) +-3.2 (PDF+alpha s) +-1.9 (PDF) +-2.6 (alpha s) ; +1.23% -1.2% (theory) +0.97% -0.99% (mq) +0.59% -0.64% (alphas)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGluHto2Mu_M130	σ [pb]	0.009150375 $= 48.75 \times (0.0001877)$	+4.5% -6.5% (theory) +-3.8% (TH Gaussian) +-3.2% (PDF+alpha s) +-1.8% (PDF) +-2.6% (alpha s) for XS ; +1.22% 1.21% (theory) +0.86% -0.82% (mQ) +0.52% -0.47% (alphas) for BR	https://arxiv.org/pdf/2402.09955forXSandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGluHToTauTau_M125	σ [pb]	3.2758656 $= 52.23 \times (0.06272)$	+4.6% -6.7% (theory) +-3.9 (TH Gaussian) +-3.2 (PDF+alpha s) +-1.9 (PDF) +-2.6 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
GluGluHToTauTau_M125_Filtered	σ [pb]	1.2762772 $= 52.23 \times (0.06272) \times 3.896e-01$	+4.6% -6.7% (theory) +-3.9 (TH Gaussian) +-3.2 (PDF+alpha s) +-1.9 (PDF) +-2.6 (alpha s) +- 0.534 (filter eff)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap*filterefffromgenXsec	—
GluGluHToWWTo2L2Nu_M125	σ [pb]	3.6364333 $= 52.23 \times (0.2137) \times (3 \times 0.1086)$	+4.6% -6.7% (theory) +-3.9 (TH Gaussian) +-3.2 (PDF+alpha s) +-1.9 (PDF) +-2.6 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
GluGluHToBB_M125	σ [pb]	30.418752 $= 52.23 \times (0.5824)$	+4.6% -6.7% (theory) +-3.9 (TH Gaussian) +-3.2 (PDF+alpha s) +-1.9 (PDF) +-2.6 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGlutoContinto2Zto2E2Nu	σ [pb]	23.31	$\pm 8.361e-05$	genXSecanalyzer	—
GluGlutoContinto2Zto2E2Mu	σ [pb]	6.115	$\pm 7.528e-05$	genXSecanalyzer	—
GluGlutoContinto2Zto2E2Tau	σ [pb]	6.242	± 0.0006294	genXSecanalyzer	—
GluGlutoContinto2Zto2Mu2Nu	σ [pb]	23.31	$\pm 7.103e-05$	genXSecanalyzer	—
GluGlutoContinto2Zto2Mu2Tau	σ [pb]	6.242	± 0.0006294	genXSecanalyzer	—
GluGlutoContinto2Zto4E	σ [pb]	3.06	± 0.004207	genXSecanalyzer	—
GluGlutoContinto2Zto4Mu	σ [pb]	3.042	± 0.003596	genXSecanalyzer	—
GluGlutoContinto2Zto4Tau	σ [pb]	3.042	± 0.003596	genXSecanalyzer	—
VBFHToBB_M125	σ [pb]	2.3750272 $= 4.078 \times (0.5824)$	+0.5% -0.3% (QCD Scale) +-2.1% (PDF alpha s) +-2.1% (PDF) +-0.5 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHto2Tau	σ [pb]	4.18	± 0.006886	XSDBVBFHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2025-09-0412:43:24	—
VBFHto2Tau_Filtered	σ [pb]	1.717	± 0.005038	XSDBVBFHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2025-09-0412:43:22	—
WminusH_Hbb_Wlnu	σ [pb]	0.10771876 $= 0.5677 \times 0.5824 \times 3 \times 0.1086$	+0.4% -0.7% (QCD scale) ±1.8% (PDF+alphas) ±1.6 (PDF) ±0.9 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
WminusH_Hbb_Wqq	σ [pb]	0.22290972 $= 0.5677 \times 0.5824 \times (1-3 \times 0.1086)$	+0.4% -0.7% (QCD scale) ±1.8% (PDF+alphas) ±1.6 (PDF) ±0.9 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
WminusHto2Tau_Filtered	σ [pb]	0.2307	± 0.0005923	XSDBWminusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8modifiedOn2025-09-0412:43:26	—

Continued on next page

Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
WminusHto2Tau	σ [pb]	0.5788	$\pm 1.752e-05$	XSDBWminusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8modifiedOn2025-09-0412:43:28	—
WplusH_Hbb_Wlnu	σ [pb]	0.16866515 $= 0.8889 \times 0.5824 \times 3 \times 0.1086$	+0.4% -0.7% (QCD scale) ±1.8% (PDF+alphas) ±1.6 (PDF) ±0.9 (alpha s)	https://indico.cern.ch/event/1119741/contributions/4715908/attachments/2383849/4073592/YR4_13p6_VH_update.pdf	—
WplusH_Hbb_Wqq	σ [pb]	0.34903021 $= 0.8889 \times 0.5824 \times (1 - 3 \times 0.1086)$	+0.4% -0.7% (QCD scale) ±1.8% (PDF+alphas) ±1.6 (PDF) ±0.9 (alpha s)	https://indico.cern.ch/event/1119741/contributions/4715908/attachments/2383849/4073592/YR4_13p6_VH_update.pdf	—
WplusHto2Tau_Filtered	σ [pb]	0.3504	±0.0009183	XSDBWplusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8modifiedOn2025-09-0412:43:29	—
WplusHto2Tau	σ [pb]	0.9273	±0.0001175	XSDBWplusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8modifiedOn2025-09-0412:43:31	—
QCD_HT40to70_MLM	σ [pb]	$3.114e+08$	±843600	XSDBQCD-4Jets_HT-40to70_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:52	—
QCD_HT70to100_MLM	σ [pb]	58500000	±167400	XSDBQCD-4Jets_HT-70to100_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:55	—
QCD_HT100to200_MLM	σ [pb]	25060000	±79510	GenXSecAnalyzer"root://cms-xrd-global.cern.ch///store/mc/Run3Summer22EEMiniAODv4/QCD-4Jets_HT-100to200_TuneCP5_13p6TeV_madgraphMLM-pythia8/MINIAODSIM/130X_mcRun3_2022_realistic_postEE_v6-v2/2520000/00527881-de92-4352-a197-c253c31c6aa7.root"	—
QCD_HT200to400_MLM	σ [pb]	1961000	±5862	XSDBQCD-4Jets_HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:48	—
QCD_HT400to600_MLM	σ [pb]	95620	±289.3	XSDBQCD-4Jets_HT-400to600_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:50	—
QCD_HT600to800_MLM	σ [pb]	13540	±41.11	XSDBQCD-4Jets_HT-600to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:54	—
QCD_HT800to1000_MLM	σ [pb]	3033	±9.206	XSDBQCD-4Jets_HT-800to1000_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:57	—
QCD_HT1000to1200_MLM	σ [pb]	883.7	±2.69	XSDBQCD-4Jets_HT-1000to1200_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:40	—
QCD_HT1200to1500_MLM	σ [pb]	383.5	±1.168	XSDBQCD-4Jets_HT-1200to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:42	—
QCD_HT1500to2000_MLM	σ [pb]	125.2	±0.381	XSDBQCD-4Jets_HT-1500to2000_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:43	—
QCD_HT2000toInf_MLM	σ [pb]	26.49	±0.0807	XSDBQCD-4Jets_HT-2000_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:45	—
QCD_PT_10to30_EMEnriched	σ [pb]	6854000	±68270	XSDBQCD_PT-10to30_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:28	—
QCD_PT_30to50_EMEnriched	σ [pb]	6698000	±62870	XSDBQCD_PT-30to50_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:12	—
QCD_PT_50to80_EMEnriched	σ [pb]	2113000	±19470	XSDBQCD_PT-50to80_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:12	—
QCD_PT_80to120_EMEnriched	σ [pb]	394100	±3654	XSDBQCD_PT-80to120_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:48	—
QCD_PT_120to170_EMEnriched	σ [pb]	71600	±330.6	GenXSecAnalyzer"root://cms-xrd-global.cern.ch///store/mc/Run3Summer22EEMiniAODv4/QCD_PT-120to170_EMEnriched_TuneCP5_13p6TeV_pythia8/MINIAODSIM/130X_mcRun3_2022_realistic_postEE_v6-v2/2520000/4ef4de22-1e29-41db-acf0-9699123caa6d.root"	—
QCD_PT_170to300_EMEnriched	σ [pb]	17920	±164.5	XSDBQCD_PT-170to300_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:33	—
QCD_PT_300_EMEnriched	σ [pb]	1231	±11.28	XSDBQCD_PT-300_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:37	—
QCD_PT_300toInf_EMEnriched	σ [pb]	1224	±11.46	XSDBQCD_PT-300toInf_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:39	—
QCD_PT_15to20_MuEnrichedPt5	σ [pb]	2960000	±29320	XSDBQCD_PT-15to20_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:11	—
QCD_PT_20to30_MuEnrichedPt5	σ [pb]	2679000	±26580	XSDBQCD_PT-20to30_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:54	—
QCD_PT_30to50_MuEnrichedPt5	σ [pb]	1465000	±14360	XSDBQCD_PT-30to50_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:41	—
QCD_PT_50to80_MuEnrichedPt5	σ [pb]	409500	±4049	XSDBQCD_PT-50to80_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:13	—
QCD_PT_80to120_MuEnrichedPt5	σ [pb]	96200	±916.9	XSDBQCD_PT-80to120_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:49	—
QCD_PT_120to170_MuEnrichedPt5	σ [pb]	22980	±215.1	XSDBQCD_PT-120to170_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:51	—
QCD_PT_170to300_MuEnrichedPt5	σ [pb]	7763	±23.67	XSDBQCD_PT-170to300_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0319:47:43	—

Continued on next page

Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
QCD_PT_300to470_MuEnrichedPt5	σ [pb]	699.1	± 6.639	XSDBQCD_PT-300to470_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:38	—
QCD_PT_470to600_MuEnrichedPt5	σ [pb]	68.24	± 0.2049	XSDBQCD_PT-470to600_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0319:47:45	—
QCD_PT_600to800_MuEnrichedPt5	σ [pb]	21.37	± 0.199	XSDBQCD_PT-600to800_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:13	—
QCD_PT_800to1000_MuEnrichedPt5	σ [pb]	3.913	± 0.03526	XSDBQCD_PT-800to1000_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:47	—
QCD_PT_1000_MuEnrichedPt5	σ [pb]	1.323	± 0.003921	XSDBQCD_PT-1000_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0319:47:38	—
QCD_PT_15to30	σ [pb]	$1.301e+09$	± 2135000	XSDBQCD_PT-15to30_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0320:23:32	—
QCD_PT_30to50	σ [pb]	$1.133e+08$	± 589300	XSDBQCD_PT-30to50_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:27	—
QCD_PT_50to80	σ [pb]	16760000	± 27450	XSDBQCD_PT-50to80_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0319:47:46	—
QCD_PT_80to120	σ [pb]	2534000	± 13280	XSDBQCD_PT-80to120_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:49	—
QCD_PT_120to170	σ [pb]	445800	± 2174	XSDBQCD_PT-120to170_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:29	—
QCD_PT_170to300	σ [pb]	113700	± 547.6	XSDBQCD_PT-170to300_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:34	—
QCD_PT_300to470	σ [pb]	7559	± 36.67	XSDBQCD_PT-300to470_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:12	—
QCD_PT_470to600	σ [pb]	626.4	± 2.863	XSDBQCD_PT-470to600_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:44	—
QCD_PT_600to800	σ [pb]	178.6	± 0.8268	XSDBQCD_PT-600to800_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:45	—
QCD_PT_800to1000	σ [pb]	30.57	± 0.1405	XSDBQCD_PT-800to1000_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:47	—
QCD_PT_1000to1400	σ [pb]	8.92	± 0.04204	XSDBQCD_PT-1000to1400_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:26	—
QCD_PT_1400to1800	σ [pb]	0.8103	± 0.003798	XSDBQCD_PT-1400to1800_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:30	—
QCD_PT_1800to2400	σ [pb]	0.1148	± 0.0005556	XSDBQCD_PT-1800to2400_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:35	—
QCD_PT_2400to3200	σ [pb]	0.007542	$\pm 3.779e-05$	XSDBQCD_PT-2400to3200_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:36	—
QCD_PT_3200	σ [pb]	0.0002331	$\pm 1.208e-06$	XSDBQCD_PT-3200_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:43	—
TbarBtoLminusNuB_s_channel_4FS	σ [pb]	1.4776306 $= 4.534 \times (1-0.6741)$	+0.059 -0.043 (scale)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef#Single_top_quark_t_channel_crosstopratiotakenfromXSDB,BRtakenfromW->LNupdg(t->Wb100%)	—
TBbartoLplusNuBbar_s_channel_4FS	σ [pb]	2.3608196 $= 7.244 \times (1-0.6741)$	+0.059 - 0.043 (scale)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef#Single_top_quark_t_channel_crosstopratiotakenfromXSDB,BRtakenfromW->LNupdg(t->Wb100%)	—
TbarBQ_t_channel_4FS	σ [pb]	87.2	+0.9 -0.8 (scale) +1.5 -1.3(PDF+alphaS) +1.8 -1.5(Total) +0.6 -0.7(mass) +0.2 -0.2 (Ebeam) +/-0.1 (Integration)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef	—
TBbarQ_t_channel_4FS	σ [pb]	145	+1.7 -1.1 (scale) +2.3 -1.5(PDF+alphaS) +2.8 -1.9(Total) +1.3 -0.9(mass) +0.4 -0.3 (Ebeam) +/-0.1 (Integration)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef	—
TbarBQtoLNu_t_channel_4FS	σ [pb]	23.34	± 0.05875	XSDBTbarBQtoLNu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8modifiedOn2025-09-2209:07:43	—
TbarBQto2Q_t_channel_4FS	σ [pb]	46.73	—	XSDB	—
TBbarQtoLNu_t_channel_4FS	σ [pb]	38.6	—	XSDB	—
TBbarQto2Q_t_channel_4FS	σ [pb]	77.26	—	XSDB	—
TbarWplusto2L2Nu	σ [pb]	4.6726331 $= 87.9 \times (0.5005) \times (1-0.6741) \times (1-0.6741)$	+2.0 -1.9 (scale) +-2.4 (PDF AlphaS) +3.1 -3.1 (Total) +1.3 -1.3 (mass) +0.2 -0.2 (Ebeam)	Valuetakenfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TbarWplusto4Q	σ [pb]	19.991326 $= 87.9 \times (0.5005) \times (0.6741) \times (0.6741)$	+2.0 -1.9 (scale) +-2.4 (PDF AlphaS) +3.1 -3.1 (Total) +1.3 -1.3 (mass) +0.2 -0.2 (Ebeam)	Valuetakenfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TbarWplustoLNu2Q	σ [pb]	19.32999 $= 87.9 \times (0.5005) \times (((1-0.6741) \times (0.6741)) + ((0.6741) \times (1-0.6741)))$	+2.0 -1.9 (scale) +-2.4 (PDF AlphaS) +3.1 -3.1 (Total) +1.3 -1.3 (mass) +0.2 -0.2 (Ebeam)	Valuetakenfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—

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Sample	Type	Value	Uncertainty	Reference	Comments
TWminusto2L2Nu	σ [pb]	4.6632971 $= 87.9 \times (0.4995) \times (1 - 0.6741) \times (1 - 0.6741)$	+2.0 -1.9 (scale) +-2.4 (PDF AlphaS) +3.1 -3.1 (Total) +1.3 -1.3 (mass) +0.2 -0.2 (Ebeam)	Valuetakenfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TWminusto4Q	σ [pb]	19.951384 $= 87.9 \times (0.4995) \times (0.6741) \times (0.6741)$	+2.0 -1.9 (scale) +-2.4 (PDF AlphaS) +3.1 -3.1 (Total) +1.3 -1.3 (mass) +0.2 -0.2 (Ebeam)	Valuetakenfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TWminustoLNu2Q	σ [pb]	19.291369 $= 87.9 \times (0.4995) \times (((1 - 0.6741) \times (0.6741)) + ((0.6741) \times (1 - 0.6741)))$	+2.0 -1.9 (scale) +-2.4 (PDF AlphaS) +3.1 -3.1 (Total) +1.3 -1.3 (mass) +0.2 -0.2 (Ebeam)	Valuetakenfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TT	σ [pb]	923.6	+22.6 -33.4 (scale) +/- 22.8 (PDF alphas) -24.6 +25.4 (mass) (to be multiplied by the BR)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/TtbarNNLO#Updated_reference_cross_sections	—
TTto2L2Nu	σ [pb]	98.096304 $= 923.6 \times (1 - 0.6741) \times (1 - 0.6741)$	+22.6 -33.4 (scale) +/- 22.8 (PDF alphas) -24.6 +25.4 (mass) (to be multiplied by the BR)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/TtbarNNLO#Updated_reference_cross_sections	—
TTto4Q	σ [pb]	419.69382 $= 923.6 \times 0.6741 \times 0.6741$	+22.6 -33.4 (scale) +/- 22.8 (PDF alphas) -24.6 +25.4 (mass) (to be multiplied by the BR)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/TtbarNNLO#Updated_reference_cross_sections	—
TTtoLNu2Q	σ [pb]	405.80987 $= 923.6 \times 0.6741 \times (1 - 0.6741) \times 2$	+22.6 -33.4 (scale) +/- 22.8 (PDF alphas) -24.6 +25.4 (mass) (to be multiplied by the BR)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/TtbarNNLO#Updated_reference_cross_sections	—
TTZ_Zto2Q	σ [pb]	0.6603	± 0.003767	XSDBTTZ-ZtoQQ-1Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:49:29	—
TTWH	σ [pb]	0.001252	$\pm 4.375e-07$	XSDBTTWH_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2024-04-0319:49:28	—
TTWW	σ [pb]	0.008203	$\pm 1.404e-05$	XSDBTTWW_TuneCP5_13p6TeV_madgraph-madspin-pythia8modifiedOn2023-05-2621:19:17	—
TTWZ	σ [pb]	0.002715	$\pm 5.963e-07$	XSDBTTWZ_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2025-09-1715:30:00	—
TTZH	σ [pb]	0.001288	$\pm 4.101e-07$	XSDBTTZH_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2024-04-0319:49:34	—
TTZH_ZHto4B	σ [pb]	0.0001961	$\pm 4.663e-08$	XSDBTTZH-ZHto4B_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2025-09-2209:07:41	—
TTZZ	σ [pb]	0.001579	$\pm 3.248e-06$	XSDBTTZZ_TuneCP5_13p6TeV_madgraph-madspin-pythia8modifiedOn2023-05-2621:19:17	—
TTZZ_ZZto4B	σ [pb]	$3.623e-05$	$\pm 1.217e-08$	XSDBTTZZ-ZZto4B_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2025-09-2209:07:41	—
TTHto2B_M125	σ [pb]	0.331968 $= 0.57 \times (0.5824)$	+6% -9.3% (QCD Scale) +-3.5% (PDF alpha s) +-3% (PDF) +-2 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
TTHtoNon2B_M125	σ [pb]	0.238032 $= 0.57 \times (1 - 0.5824)$	+6% -9.3% (QCD Scale) +-3.5% (PDF alpha s) +-3% (PDF) +-2 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
TTH_Hto2Mu	σ [pb]	0.000124032 $= 0.57 \times (0.0002176)$	+6% -9.3% (QCD Scale) +-3.5% (PDF alpha s) +-3% (PDF) +-2 (alpha s) ; +1.23% -1.2% (theory) +0.97% -0.99% (mq) +0.59% -0.64% (alphas)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
VBFHto2Mu_M120	σ [pb]	0.0009880994 $= 4.078 \times (0.0002423)$	+0.5% -0.3% (QCD Scale) +-2.1% (PDF alpha s) +-2.1% (PDF) +-0.5 (alpha s) ; +1.23% 1.23% (theory) +1.07% -1.09% (mQ) +0.69% -0.70% (alphas) for BR	https://arxiv.org/pdf/2402.09955forXSandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHto2Mu_M125	σ [pb]	0.0008873728 $= 4.078 \times (0.0002176)$	+0.5% -0.3% (QCD Scale) +-2.1% (PDF alpha s) +-2.1% (PDF) +-0.5 (alpha s) ; +1.23% -1.2% (theory) +0.97% -0.99% (mq) +0.59% -0.64% (alphas)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHto2Mu_M130	σ [pb]	0.0007654406 $= 4.078 \times (0.0001877)$	+0.5% -0.3% (QCD Scale) +-2.1% (PDF alpha s) +-2.1% (PDF) +-0.5 (alpha s) ; +1.22% 1.21% (theory) +0.86% -0.82% (mQ) +0.52% -0.47% (alphas) for BR	https://arxiv.org/pdf/2402.09955forXSandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHto2Mu	σ [pb]	0.0008873728 $= 4.078 \times (0.0002176)$	+0.5% -0.3% (QCD Scale) +-2.1% (PDF alpha s) +-2.1% (PDF) +-0.5 (alpha s) ; +1.23% -1.2% (theory) +0.97% -0.99% (mq) +0.59% -0.64% (alphas)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHToTauTau_M125	σ [pb]	0.25577216 $= 4.078 \times (0.06272)$	+0.5% -0.3% (QCD Scale) +-2.1% (PDF alpha s) +-2.1% (PDF) +-0.5 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
VBFHToWWTo2L2Nu_M125	σ [pb]	0.28392447 $= 4.078 \times (0.2137) \times (3 \times 0.1086)$	+0.5% -0.3% (QCD Scale) +-2.1% (PDF alpha s) +-2.1% (PDF) +-0.5 (alpha s)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
WminusH_Hto2Mu_WtoAll	σ [pb]	0.00012353152 $= 0.5677 \times (0.0002176)$	+0.4% -0.7% (QCD scale) ±1.8% (PDF+alphas) ±1.6 (PDF) ±0.9 (alpha s) ; +1.23% -1.2% (theory) +0.97% -0.99% (mq) +0.59% -0.64% (alphas)	https://indico.cern.ch/event/1119741/contributions/4715908/attachments/2383849/4073592/YR4_13p6_VH_update.pdf	—

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Sample	Type	Value	Uncertainty	Reference	Comments
WplusH_Hto2Mu_WtoAll	σ [pb]	0.00019342464 = 0.8889 × (0.0002176)	+0.4% -0.7% (QCD scale) ±1.8% (PDF+alphas) ±1.6 (PDF) ±0.9 (alpha s) ; +1.23% -1.2% (theory) +0.97% -0.99% (mq) +0.59% -0.64% (alphas)	https://indico.cern.ch/event/1119741/contributions/4715908/attachments/2383849/4073592/YR4_13p6_VH_update.pdf	—
WtoLNu_0J_amcatnloFXFX	σ [pb]	55760	±75.39	XSDBWtoLNu-2Jets_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:09:44	—
WtoLNu_1J_amcatnloFXFX	σ [pb]	9529	±56.7	XSDBWtoLNu-2Jets_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:09:46	—
WtoLNu_1J_madgraphMLM	σ [pb]	9084	±24.69	XSDBWtoLNu-4Jets_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:09	—
WtoLNu_2J_amcatnloFXFX	σ [pb]	3532	±34.81	XSDBWtoLNu-2Jets_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:09:47	—
WtoLNu_2J_madgraphMLM	σ [pb]	2925	±8.555	XSDBWtoLNu-4Jets_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:12	—
WtoLNu_3J_madgraphMLM	σ [pb]	861.7	±2.624	XSDBWtoLNu-4Jets_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:13	—
WtoLNu_4J_madgraphMLM	σ [pb]	416.5	±1.28	XSDBWtoLNu-4Jets_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:17	—
WtoLNu_NNLO_QCD_NLO_EW_one_flav	σ [pb]	21141.7	+1.2%% -1.1%% +-0.8%% (e-nu) +1.1%% -1.4%% +-0.7%% (e+nu)	https://twiki.cern.ch/twiki/bin/viewauth/CMS/MATRIXCrossSectionsat13p6TeV	—
WtoLNu_NNLO_QCD+NLO_EW	σ [pb]	63425.1	+1.2%% -1.1%% +-0.8%% (e-nu) +1.1%% -1.4%% +-0.7%% (e+nu)	https://twiki.cern.ch/twiki/bin/viewauth/CMS/MATRIXCrossSectionsat13p6TeV	—
WtoLNu_MLNu_0to120_HT_40To100_madgraph	σ [pb]	4254	±12.08	XSDBWtoLNu-4Jets_MLNu-0to120_HT-40to100_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:25	—
WtoLNu_MLNu_0to120_HT_100To400_madgraph	σ [pb]	1626	±4.823	XSDBWtoLNu-4Jets_MLNu-0to120_HT-100to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:18	—
WtoLNu_MLNu_0to120_HT_400To800_madgraph	σ [pb]	59.99	±0.1803	XSDBWtoLNu-4Jets_MLNu-0to120_HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:23	—
WtoLNu_MLNu_0to120_HT_800To1500_madgraph	σ [pb]	6.23	±0.01876	XSDBWtoLNu-4Jets_MLNu-0to120_HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:28	—
WtoLNu_MLNu_0to120_HT_1500To2500_madgraph	σ [pb]	0.4477	±0.00135	XSDBWtoLNu-4Jets_MLNu-0to120_HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:20	—
WtoLNu_PTLNu_40to100_1J_amcatnloFXFX	σ [pb]	4427	±28.3	XSDBWtoLNu-2Jets_PTLNu-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_40to100_2J_amcatnloFXFX	σ [pb]	1598	±16.36	XSDBWtoLNu-2Jets_PTLNu-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_100to200_1J_amcatnloFXFX	σ [pb]	368.2	±1.926	XSDBWtoLNu-2Jets_PTLNu-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_100to200_2J_amcatnloFXFX	σ [pb]	421.9	±3.533	XSDBWtoLNu-2Jets_PTLNu-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_200to400_1J_amcatnloFXFX	σ [pb]	25.6	±0.1207	XSDBWtoLNu-2Jets_PTLNu-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_200to400_2J_amcatnloFXFX	σ [pb]	54.77	±0.3764	XSDBWtoLNu-2Jets_PTLNu-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_400to600_1J_amcatnloFXFX	σ [pb]	0.8785	±0.003223	XSDBWtoLNu-2Jets_PTLNu-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_400to600_2J_amcatnloFXFX	σ [pb]	3.119	±0.01749	XSDBWtoLNu-2Jets_PTLNu-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_600_1J_amcatnloFXFX	σ [pb]	0.1053	±0.0003437	XSDBWtoLNu-2Jets_PTLNu-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_600_2J_amcatnloFXFX	σ [pb]	0.5261	±0.002417	XSDBWtoLNu-2Jets_PTLNu-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WToMuNu_M_200	σ [pb]	7.693	±0.0408	XSDBWtoMuNu_M-200_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:05:35	—
WToTauNu_M_200	σ [pb]	7.753	±0.04088	XSDBWtoNuTau_M-200_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:05:40	—
WW	σ [pb]	80.23	±0.3733	XSDBWW_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:02:32	—
WWto2L2Nu	σ [pb]	11.79	±0.004216	XSDBWWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:06:05	—
WWW	σ [pb]	0.2328	±0.0001247	XSDBWWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8modifiedOn2024-03-0204:47:30	—
WWZ	σ [pb]	0.1851	$\pm 9.482e-05$	XSDBWWZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8modifiedOn2023-05-2621:19:27	—
WZZ	σ [pb]	0.06206	$\pm 3.689e-05$	XSDBWZZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8modifiedOn2023-05-2621:19:28	—
WZ	σ [pb]	29.1	±0.1318	XSDBWZ_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:02:34	—

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Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
WZto2L2Q_powheg	σ [pb]	7.568	$\pm$0.003908	XSDBWZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:06:15	—
WZto3LNu_amcatnloFXFX	σ [pb]	5.315	$\pm$0.01577	XSDBWZto3LNu-1Jets-4FS_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:06:16	—
WZtoL3Nu_powheg	σ [pb]	4.924	$\pm$0.00237	XSDBWZto3LNu_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:06:18	—
WZto3LNu_powheg	σ [pb]	4.924	$\pm$0.00237	XSDBWZto3LNu_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2025-09-19	—
WZtoL3Nu_amcatnloFXFX	σ [pb]	3.445	$\pm$0.01204	XSDBWZtoL3Nu-1Jets-4FS_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:06:23	—
WZtoLNu2Q_amcatnloFXFX	σ [pb]	12.11	$\pm$0.04269	XSDBWZtoLNu2Q-1Jets-4FS_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:06:25	—
WZtoLNu2Q_powheg	σ [pb]	15.87	$\pm$0.007874	XSDBWZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:06:26	—
WWtoLNu2Q	σ [pb]	48.94	$\pm$0.0175	XSDBWWtoLNu2Q_TuneCP5_13p6TeVmodifiedOn2024-04-0320:06:13	—
WWto4Q-1Jets	σ [pb]	54.82	$\pm$0.1846	XSDBWWto4Q-1Jets-4FS_TuneCP5_13p6TeVmodifiedOn2024-04-0320:06:06	—
WWto4Q	σ [pb]	50.79	$\pm$0.01816	XSDBWWto4Q_TuneCP5_13p6TeVmodifiedOn2024-04-0320:06:10	—
WWtoLNu2Q-1Jets	σ [pb]	52.98	$\pm$0.1795	XSDBWWtoLNu2Q-1Jets-4FS_TuneCP5_13p6TeVmodifiedOn2024-04-0320:06:12	—
ZH_Hbb_Zll	σ [pb]	0.05550817 $= 9.439e-1 \times 0.5824 \times 3 \times 0.033658$	+3.7 -3.2 (qcd scale) +-1.6 (pdf alpha) +-1.3 (pdf) +- 0.9 (alpha)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
ZH_Hbb_Zqq	σ [pb]	0.38431989 $= 9.439e-1 \times 0.5824 \times 0.69911$	+3.7 -3.2 (qcd scale) +-1.6 (pdf alpha) +-1.3 (pdf) +- 0.9 (alpha)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
ZH_Hto2Mu_ZtoAll	σ [pb]	0.00020539264 $= 0.9439 \times (0.0002176)$	+3.7 -3.2 (qcd scale) +-1.6 (pdf alpha) +-1.3 (pdf) +- 0.9 (alpha) ; +1.23% -1.2% (theory) +0.97% -0.99% (mq) +0.59% -0.64% (alphas)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
ZHto2Tau	σ [pb]	0.059201408 $= 0.9439 \times 0.06272$	+3.7 -3.2 (qcd scale) +-1.6 (pdf alpha) +-1.3 (pdf) +- 0.9 (alpha) ; +1.17% -1.16% (theory) +0.98% -0.99% (mq) +0.62% -0.62% (alphas)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandBRfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBR	—
ZHto2Tau_Filtered	σ [pb]	0.023277994 $= 0.9439 \times 0.06272 \times 0.3932$	+3.7 -3.2 (qcd scale) +-1.6 (pdf alpha) +-1.3 (pdf) +- 0.9 (alpha) ; +1.17% -1.16% (theory) +0.98% -0.99% (mq) +0.62% -0.62% (alphas) ; 0.8% (filter eff)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandBRfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBR*genXsecanalyzerfiltereff	—
Zto2Nu_HT_100To200	σ [pb]	273.7	$\pm$0.8085	XSDBZto2Nu-4Jets_HT-100to200_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:13:00	—
Zto2Nu_HT_200To400	σ [pb]	75.96	$\pm$0.2275	XSDBZto2Nu-4Jets_HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:13:04	—
Zto2Nu_HT_400To800	σ [pb]	13.19	$\pm$0.03963	XSDBZto2Nu-4Jets_HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:13:07	—
Zto2Nu_HT_800To1500	σ [pb]	1.364	$\pm$0.004118	XSDBZto2Nu-4Jets_HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:13:09	—
Zto2Nu_HT_1500To2500	σ [pb]	0.09865	$\pm$0.0002974	XSDBZto2Nu-4Jets_HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:13:02	—
Zto2Nu_HT_2500ToInf	σ [pb]	0.006699	$\pm 2.019e-05$	XSDBZto2Nu-4Jets_HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:13:06	—
Zto2Q_HT_100to400	σ [pb]	6328	$\pm$17.95	XSDBZto2Q-4Jets_Bin-HT-100to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2025-09-2209:08:33	—
Zto2Q_HT_400to800	σ [pb]	145.1	$\pm$0.4362	XSDBZto2Q-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2025-09-2209:08:33	—
Zto2Q_HT_800to1500	σ [pb]	12.9	$\pm$0.01236	XSDBZto2Q-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2026-01-2520:47:07	—
Zto2Q_HT_1500to2500	σ [pb]	0.8541	$\pm 5.449e-05$	XSDBZto2Q-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2026-01-2520:47:08	—
Zto2Q_HT_2500toInf	σ [pb]	0.05684	$\pm$0.0008198	XSDBZto2Q-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2026-01-2520:47:08	—
Zto2Q_HT_200to400	σ [pb]	1082	$\pm$9.886	XSDBZto2Q-4Jets_HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-05-2621:19:45	—
Zto2Q_HT_400to600	σ [pb]	124.1	$\pm$1.163	XSDBZto2Q-4Jets_HT-400to600_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-05-2621:19:45	—
Zto2Q_HT_600to800	σ [pb]	27.28	$\pm$0.2549	XSDBZto2Q-4Jets_HT-600to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-05-2621:19:46	—
Zto2Q_HT_800toInf	σ [pb]	14.57	$\pm$0.1378	XSDBZto2Q-4Jets_HT-800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-05-2621:19:46	—
ZZ	σ [pb]	12.75	$\pm$0.0649	XSDBZZ_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:05:37	—

Continued on next page

Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
ZZto2L2Nu_powheg	σ [pb]	1.031	± 0.0005268	XSDBZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:06:52	—
ZZto2L2Q_amcatnloFXFX	σ [pb]	3.86	± 0.01259	XSDBZZto2L2Q-1Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:12:33	—
ZZto2L2Q_powheg	σ [pb]	6.788	± 0.003501	XSDBZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:08:04	—
ZZto2Nu2Q_powheg	σ [pb]	4.826	± 0.001296	XSDBZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:08:39	—
ZZto4L_powheg	σ [pb]	1.39	± 0.0007001	XSDBZZto4L_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:11:16	—
ZZZ	σ [pb]	0.01591	$\pm 7.828e-06$	XSDBZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8modifiedOn2023-05-2621:19:33	—
DYto2L_2Jets_EWK_MLL_50_LO	σ [pb]	13.07	± 0.003645	XSDBDYto2L-2Jets-EWK_MLL-50_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2026-01-2600:01:51	—
EWK_2L2J_MLL50_MJJ120	σ [pb]	1.418	± 0.002142	fromgridpackgenerationlogtakenfro/cvmfs/cms.cern.ch/phys_generator/gridpacks/RunIII/13p6TeV/slc7_amd64_gcc10/MadGraph5_aMCatNLO/LLJJ_50-Inf_EWK_SM_5f_LO_slc7_amd64_gcc10_CMSSW_12_4_8_tarball.tar.xzandwiththissetting	—
EWK_2Mu2J_MLL105to160_MJJ120	σ [pb]	0.0635	± 0.0003071	fromgridpackgenerationlogtakenfrom/cvmfs/cms.cern.ch/phys_generator/gridpacks/RunIII/13p6TeV/slc7_amd64_gcc10/MadGraph5_aMCatNLO/LLJJ_105-160_EWK_SM_5f_LO_slc7_amd64_gcc10_CMSSW_12_4_8_tarball.tar.xzandwiththissetting	—
TTLNu_EWK	σ [pb]	0.01769	$\pm 5.69e-06$	XSDBTTLNu-EWK_TuneCP5_13p6TeV_amcatnlo-pythia8modifiedOn2024-12-2008:23:11	—
WZto3LNu_2Jets_EWK_QCD	σ [pb]	0.5437	± 0.0001215	XSDBZto3LNu-2Jets_EWK-QCD_TuneCP5_13p6TeV_madgraph-madspin-pythia8modifiedOn2024-12-2008:28:09	—
WmWmJJ_EWK	σ [pb]	0.000122	$\pm 2.832e-05$	XSDBWmWmJJ-EWK_TuneCP5_13p6TeV-powheg-pythia8modifiedOn2025-09-2209:07:36	—

Table 2: Correction payloads, paths, and tags used in the analysis.

Era	Correction	Payload / path	Key / tag
Run3_2022	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer22_22Sep2023_V4_MC; JEC data: Summer22_22Sep2023_V4_DATA; JER: Summer22_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22CDSep23-Summer22-NanoAODv12/latest/puWeights.json.gz	Collisions2022_355100_357900_eraBCD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22CDSep23-Summer22-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoAODv12/latest/jetvetomaps.json.gz	Summer22_23Sep2023_RunCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22CDSep23-Summer22-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2022EE	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer22EE_22Sep2023_V4_MC; JEC data: Summer22EE_22Sep2023_Run{}_V4_DATA, Summer22EE_22Sep2023_V4_DATA; JER: Summer22EE_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/puWeights.json.gz	Collisions2022_359022_362760_eraEFG_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/jetvetomaps.json.gz	Summer22EE_23Sep2023_RunEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2023	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer23Prompt23_V4_MC; JEC data: Summer23Prompt23_Run{}_V4_DATA, Summer23Prompt23_V4_DATA; JER: Summer23Prompt23_RunCv1234_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23CSep23-Summer23-NanoAODv12/latest/puWeights.json.gz	Collisions2023_366403_369802_eraBC_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23CSep23-Summer23-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoAODv12/latest/jetvetomaps.json.gz	Summer23Prompt23_RunC_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23CSep23-Summer23-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2023BPix	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer23BPixPrompt23_V4_MC; JEC data: Summer23BPixPrompt23_Run{}_V4_DATA, Summer23BPixPrompt23_V4_DATA; JER: Summer23BPixPrompt23_RunD_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/puWeights.json.gz	Collisions2023_369803_370790_eraD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jetvetomaps.json.gz	Summer23BPixPrompt23_RunD_V1

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Table 2 – continued

Era	Correction	Payload / path	Key / tag
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2024	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt24_V5_MC; JEC data: Summer24Prompt24_V5_DATA; JER: Summer24Prompt24_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/puWeights_BCDEFGHI.json.gz	Collisions24_BCDEFGHI_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing_
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt24_RunBCDEFGHI_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2024_378981_386951_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2025	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt25_V3_MC; JEC data: Summer24Prompt25_V3_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/afs/cern.ch/work/v/vdamante/Hmm_newSkim/corrections/data/MUO/SF/Run3-25Prompt-Summer24-NanoAODv15/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt25_RunCDEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-25Prompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2025_391658_398903_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2026	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt26_V2_MC; JEC data: Summer24Prompt26_Run{}_V1_DATA, Summer24Prompt26_V1_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-26Prompt-Summer24-NanoAODv15/latest/muon_Z.json.gz	
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt26_RunBCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-26Prompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2026_401624_403937_golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium

Table 3: NanoAOD 2022 datasets used in the analysis

Sample	NanoAOD dataset
Data (6 samples)	
DoubleMuon_Run2022C	/DoubleMuon/Run2022C-22Sep2023-v1/NANOAOD
Muon_Run2022C	/Muon/Run2022C-22Sep2023-v1/NANOAOD
Muon_Run2022D	/Muon/Run2022D-22Sep2023-v1/NANOAOD
MuonEG_Run2022C	/MuonEG/Run2022C-22Sep2023-v1/NANOAOD
MuonEG_Run2022D	/MuonEG/Run2022D-22Sep2023-v1/NANOAOD
SingleMuon_Run2022C	/SingleMuon/Run2022C-22Sep2023-v1/NANOAOD
Simulation (107 samples)	
GluGluHto2Mu_M120	/GluGluHto2Mu_M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
GluGluHto2Mu	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
GluGluHto2Mu_amcatnlo	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
GluGluHto2Mu_MiNNLO	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powhegMiNNLO-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
GluGluHto2Mu_tuneDown	/GluGluHto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
GluGluHto2Mu_tuneUp	/GluGluHto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
GluGluHto2Mu_M130	/GluGluHto2Mu_M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
VBFHto2Mu_M125_powheg	/VBFHto2Mu_M-125_TuneCP5_withDipoleRecoil_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
ZH_Hto2Mu	/ZH_Hto2Mu_ZtoAll_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
WplusH_Hto2Mu	/WplusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
WminusH_Hto2Mu	/WminusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
TTH_Hto2Mu	/TTH_Hto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
VBFHto2Mu_M120_amcatnlo	/VBFH_Hto2Mu_M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
VBFHto2Mu_m125_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
VBFHto2Mu_M130_amcatnlo	/VBFH_Hto2Mu_M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
DYto2Mu_MLL_105to160_amcatnloFXFX	/DYto2Mu-2Jets_MLL-105To160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
DYto2L_M_10to50_amcatnloFXFX	/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v4/NANOAODSIM
DYto2L_M_50_0J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
DYto2L_M_50_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
DYto2L_M_50_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
DYto2L_M_50_amcatnloFXFX	/DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v5/NANOAODSIM
DYto2L_M_50_PTLL_40to100_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v1/NANOAODSIM
DYto2L_M_50_PTLL_40to100_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v1/NANOAODSIM
DYto2L_M_50_PTLL_100to200_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v1/NANOAODSIM
DYto2L_M_50_PTLL_100to200_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v1/NANOAODSIM
DYto2L_M_50_PTLL_200to400_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_200to400_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_400to600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_400to600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
EWK_2L2J_madgraph_herwig	/EWK_2L2J_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK_2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_M2Mu-105to160_M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
ggZH_Hto2B_Zto2Q	/ggZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM /ggZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v3/NANOAODSIM
ggZH_Hto2B_Zto2L	/ggZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM /ggZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v3/NANOAODSIM
GluGluHto2B_M125	/GluGluHto2B_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM

Continued on next page

Table 3: NanoAOD 2022 datasets used in the analysis (continued)

Sample	NanoAOD dataset
GluGluHto2Tau_M125	/GluGluHToTauTau_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_Filtered	/GluGluHTo2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_UnFiltered	/GluGluHTo2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TTHto2B_M125	/TTHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
TTHtoNon2B_M125	/TTHtoNon2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v4/NANOAODSIM
VBFHto2B_M125	/VBFHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
VBFHto2Tau_M125	/VBFHToTauTau_M125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_UnFiltered	/VBFHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_Filtered	/VBFHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WminusH_Hto2B_WtoLNu	/WminusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WminusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WplusH_Hto2B_WtoLNu	/WplusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WplusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_Filtered	/WminusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_UnFiltered	/WminusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_Filtered	/WplusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_UnFiltered	/WplusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
ZH_Hto2B_Zto2L	/ZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /ZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
ZH_Hto2B_Zto2Q	/ZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM /ZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v3/NANOAODSIM
WminusH_Hto2B_Wto2Q	/WminusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WminusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WplusH_Hto2B_Wto2Q	/WplusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WplusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TBbartoLplusNuBbar_s_channel_4FS	/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TbarWplustoLNu2Q	/TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TWminusto2L2Nu	/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TWminusto4Q	/TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TWminustoLNu2Q	/TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TTtoLNu2Q	/TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TTZ_Zto2Q	/TTZ-ZtoQQ-1Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TbarBQ_t_channel_4FS	/TbarBQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TBbarQ_t_channel_4FS	/TBbarQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TT	/TT_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /TT_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
TTZH	/TTZH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
TTZZ	/TTZZ_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WtoLNu_1J_madgraphMLM	/WtoLNu-4Jets_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v1/NANOAODSIM
WtoLNu_2J_amcatnloFXFX	/WtoLNu-2Jets_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM

Continued on next page

Table 3: NanoAOD 2022 datasets used in the analysis (continued)

Sample	NanoAOD dataset
WtoLNu_2J_madgraphMLM	/WtoLNu-4Jets_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v1/NANOAODSIM
WtoLNu_3J_madgraphMLM	/WtoLNu-4Jets_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WtoLNu_4J_madgraphMLM	/WtoLNu-4Jets_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WtoLNu_0J_amcatnloFXFX	/WtoLNu-2Jets_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM
WtoLNu_1J_amcatnloFXFX	/WtoLNu-2Jets_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WtoLNu_amcatnloFXFX	/WtoLNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WtoLNu_madgraphMLM	/WtoLNu-4Jets_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WtoLNu-4Jets_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WWtoLNu2Q_powheg	/WWtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WWtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WZto3LNu_powheg	/WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v3/NANOAODSIM /WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v3/NANOAODSIM
WZtoLNu2Q_powheg	/WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
ZZto2L2Nu_powheg	/ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
ZZto2Nu2Q_powheg	/ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM /ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5_ext1-v2/NANOAODSIM
WWW_4F	/WWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WWZ_4F	/WWZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
WZZ	/WZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
ZZZ	/ZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_Filtered	/ZHto2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_UnFiltered	/ZHto2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-v2/NANOAODSIM

Table 4: NanoAOD 2022EE datasets used in the analysis

Sample	NanoAOD dataset
Data (3 samples)	
Muon_Run2022E	/Muon/Run2022E-22Sep2023-v1/NANOAOD
Muon_Run2022F	/Muon/Run2022F-22Sep2023-v2/NANOAOD
Muon_Run2022G	/Muon/Run2022G-22Sep2023-v1/NANOAOD
Simulation (104 samples)	
GluGluHto2Mu_M120	/GluGluHto2Mu_M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2Mu	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v1/NANOAODSIM
GluGluHto2Mu_amcatnlo	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2Mu_MiNNLO	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powhegMiNNLO-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2Mu_tuneDown	/GluGluHto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2Mu_tuneUp	/GluGluHto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2Mu_M130	/GluGluHto2Mu_M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
VBFHto2Mu_M125_powheg	/VBFHto2Mu_M-125_TuneCP5_withDipoleRecoil_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
ZH_Hto2Mu	/ZH_Hto2Mu_ZtoAll_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
WplusH_Hto2Mu	/WplusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
WminusH_Hto2Mu	/WminusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
TTH_Hto2Mu	/TTH_Hto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
VBFHto2Mu_M120_amcatnlo	/VBFH_Hto2Mu_M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
VBFHto2Mu_m125_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
VBFHto2Mu_M130_amcatnlo	/VBFH_Hto2Mu_M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
DYto2Mu_MLL_105to160_amcatnloFXFX	/DYto2Mu-2Jets_MLL-105To160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
DYto2L_M_10to50_amcatnloFXFX	/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v5/NANOAODSIM
DYto2L_M_50_0J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
DYto2L_M_50_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
DYto2L_M_50_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
DYto2L_M_50_amcatnloFXFX	/DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v5/NANOAODSIM
DYto2L_M_50_PTLL_40to100_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v1/NANOAODSIM
DYto2L_M_50_PTLL_40to100_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
DYto2L_M_50_PTLL_100to200_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v1/NANOAODSIM
DYto2L_M_50_PTLL_100to200_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
DYto2L_M_50_PTLL_200to400_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v1/NANOAODSIM
DYto2L_M_50_PTLL_200to400_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
DYto2L_M_50_PTLL_400to600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v1/NANOAODSIM
DYto2L_M_50_PTLL_400to600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
DYto2L_M_50_PTLL_600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
DYto2L_M_50_PTLL_600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v1/NANOAODSIM
EWK_2L2J_madgraph_herwig	/EWK_2L2J_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK_2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_M2Mu-105to160_M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
ggZH_Hto2B_Zto2Q	/ggZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
ggZH_Hto2B_Zto2L	/ggZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2B_M125	/GluGluHto2B_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_Filtered	/GluGluHTo2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_UnFiltered	/GluGluHTo2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TTHto2B_M125	/TTHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
TTHtoNon2B_M125	/TTHtoNon2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
VBFHto2B_M125	/VBFHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_UnFiltered	/VBFHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_Filtered	/VBFHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM

Continued on next page

Table 4: NanoAOD 2022EE datasets used in the analysis (continued)

Sample	NanoAOD dataset
WminusH_Hto2B_WtoLNu	/WminusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WplusH_Hto2B_WtoLNu	/WplusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_Filtered	/WminusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_UnFiltered	/WminusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_Filtered	/WplusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_UnFiltered	/WplusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
ZH_Hto2B_Zto2L	/ZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
ZH_Hto2B_Zto2Q	/ZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WminusH_Hto2B_Wto2Q	/WminusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WplusH_Hto2B_Wto2Q	/WplusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TBbartoLplusNuBbar_s_channel_4FS	/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
TbarWplustoLNu2Q	/TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
TWminusto2L2Nu	/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
TWminusto4Q	/TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
TWminustoLNu2Q	/TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TTtoLNu2Q	/TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TTZ_Zto2Q	/TTZ-ZtoQQ-1Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TbarBQ_t_channel_4FS	/TbarBQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TBbarQ_t_channel_4FS	/TBbarQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
TTZH	/TTZH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
TTZZ	/TTZZ_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
WtoLNu_1J_madgraphMLM	/WtoLNu-4Jets_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WtoLNu_2J_amcatnloFXFX	/WtoLNu-2Jets_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WtoLNu_2J_madgraphMLM	/WtoLNu-4Jets_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WtoLNu_3J_madgraphMLM	/WtoLNu-4Jets_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WtoLNu_4J_madgraphMLM	/WtoLNu-4Jets_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WtoLNu_0J_amcatnloFXFX	/WtoLNu-2Jets_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v3/NANOAODSIM
WtoLNu_1J_amcatnloFXFX	/WtoLNu-2Jets_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WtoLNu_amcatnloFXFX	/WtoLNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WtoLNu_madgraphMLM	/WtoLNu-4Jets_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /WtoLNu-4Jets_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
WWtoLNu2Q_powheg	/WWtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /WWtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
WZto3LNu_powheg	/WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM

Continued on next page

Table 4: NanoAOD 2022EE datasets used in the analysis (continued)

Sample	NanoAOD dataset
WZtoLNu2Q_powheg	/WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
ZZto2L2Nu_powheg	/ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
ZZto2Nu2Q_powheg	/ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM /ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6_ext1-v2/NANOAODSIM
WWW_4F	/WWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WWZ_4F	/WWZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
WZZ	/WZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
ZZZ	/ZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v2/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_Filtered	/ZHto2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v4/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_UnFiltered	/ZHto2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer22EENanoAODv12-130X_mcRun3_2022_realistic_postEE_v6-v4/NANOAODSIM

Table 5: NanoAOD 2023 datasets used in the analysis

Sample	NanoAOD dataset
Data (8 samples)	
Muon0_Run2023C_v1	/Muon0/Run2023C-22Sep2023_v1-v1/NANOAOD
Muon0_Run2023C_v2	/Muon0/Run2023C-22Sep2023_v2-v1/NANOAOD
Muon0_Run2023C_v3	/Muon0/Run2023C-22Sep2023_v3-v1/NANOAOD
Muon0_Run2023C_v4	/Muon0/Run2023C-22Sep2023_v4-v1/NANOAOD
Muon1_Run2023C_v1	/Muon1/Run2023C-22Sep2023_v1-v1/NANOAOD
Muon1_Run2023C_v2	/Muon1/Run2023C-22Sep2023_v2-v1/NANOAOD
Muon1_Run2023C_v3	/Muon1/Run2023C-22Sep2023_v3-v1/NANOAOD
Muon1_Run2023C_v4	/Muon1/Run2023C-22Sep2023_v4-v2/NANOAOD
Simulation (104 samples)	
GluGluHto2Mu_M120	/GluGluHto2Mu_M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
GluGluHto2Mu	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
GluGluHto2Mu_amcatnlo	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
GluGluHto2Mu_MiNNLO	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powhegMiNNLO-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
GluGluHto2Mu_tuneDown	/GluGluHto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
GluGluHto2Mu_tuneUp	/GluGluHto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
GluGluHto2Mu_M130	/GluGluHto2Mu_M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
VBFHto2Mu_M125_powheg	/VBFHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
ZH_Hto2Mu	/ZH_Hto2Mu_ZtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
WplusH_Hto2Mu	/WplusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
WminusH_Hto2Mu	/WminusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
TTH_Hto2Mu	/TTH_Hto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
VBFHto2Mu_M120_amcatnlo	/VBFH_Hto2Mu_M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
VBFHto2Mu_m125_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
VBFHto2Mu_M130_amcatnlo	/VBFH_Hto2Mu_M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
DYto2Mu_MLL_105to160_amcatnloFXFX	/DYto2Mu-2Jets_MLL-105To160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
DYto2L_M_10to50_amcatnloFXFX	/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOAODSIM /DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14_ext1-v4/NANOAODSIM
DYto2L_M_50_0J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOAODSIM
DYto2L_M_50_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOAODSIM
DYto2L_M_50_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOAODSIM
DYto2L_M_50_amcatnloFXFX	/DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
DYto2L_M_50_PTLL_40to100_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v1/NANOAODSIM
DYto2L_M_50_PTLL_40to100_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v1/NANOAODSIM
DYto2L_M_50_PTLL_100to200_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v1/NANOAODSIM
DYto2L_M_50_PTLL_100to200_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v1/NANOAODSIM
DYto2L_M_50_PTLL_200to400_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_200to400_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_400to600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_400to600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOAODSIM
EWK_2L2J_madgraph_herwig	/EWK_2L2J_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK_2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_M2Mu-105to160_M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
ggZH_Hto2B_Zto2Q	/ggZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
ggZH_Hto2B_Zto2L	/ggZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM

Continued on next page

Table 5: NanoAOD 2023 datasets used in the analysis (continued)

Sample	NanoAOD dataset
GluGluHto2B_M125	/GluGluHto2B_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_Filtered	/GluGluHTo2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_UnFiltered	/GluGluHTo2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
TTHto2B_M125	/TTHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
TTHtoNon2B_M125	/TTHtoNon2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
VBFHto2B_M125	/VBFHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
VBFHto2Tau_M125	/VBFHToTauTau_M125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_UnFiltered	/VBFHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_Filtered	/VBFHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
WminusH_Hto2B_WtoLNu	/WminusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WplusH_Hto2B_WtoLNu	/WplusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_Filtered	/WminusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_UnFiltered	/WminusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_Filtered	/WplusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_UnFiltered	/WplusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
ZH_Hto2B_Zto2L	/ZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
ZH_Hto2B_Zto2Q	/ZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WminusH_Hto2B_Wto2Q	/WminusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WplusH_Hto2B_Wto2Q	/WplusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
TBbartoLplusNuBbar_s_channel_4FS	/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v4/NANOAODSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v4/NANOAODSIM
TbarWplustoLNu2Q	/TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v6/NANOAODSIM
TWminusto2L2Nu	/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
TWminusto4Q	/TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
TWminustoLNu2Q	/TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
TTtoLNu2Q	/TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v3/NANOAODSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
TbarBQ_t_channel_4FS	/TbarBQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
TBbarQ_t_channel_4FS	/TBbarQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
TTZH	/TTZH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v3/NANOAODSIM
TTZZ	/TTZZ_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v4/NANOAODSIM
WtoLNu_1J_madgraphMLM	/WtoLNu-4Jets_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WtoLNu_2J_amcatnloFXFX	/WtoLNu-2Jets_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WtoLNu_2J_madgraphMLM	/WtoLNu-4Jets_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WtoLNu_3J_madgraphMLM	/WtoLNu-4Jets_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WtoLNu_4J_madgraphMLM	/WtoLNu-4Jets_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WtoLNu_0J_amcatnloFXFX	/WtoLNu-2Jets_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WtoLNu_1J_amcatnloFXFX	/WtoLNu-2Jets_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WtoLNu_amcatnloFXFX	/WtoLNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM
WtoLNu_madgraphMLM	/WtoLNu-4Jets_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOAODSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOAODSIM
WWtoLNu2Q_powheg	/WWtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM

Continued on next page

Table 5: NanoAOD 2023 datasets used in the analysis (continued)

Sample	NanoAOD dataset
WZto3LNu_powheg	/WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOAODSIM /WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOAODSIM
WZtoLNu2Q_powheg	/WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
ZZto2L2Nu_powheg	/ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
ZZto2Nu2Q_powheg	/ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOAODSIM
WWW_4F	/WWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WWZ_4F	/WWZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
WZZ	/WZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
ZZZ	/ZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_Filtered	/ZHto2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v5/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_UnFiltered	/ZHto2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v5/NANOAODSIM

Table 6: NanoAOD 2023BPix datasets used in the analysis

Sample	NanoAOD dataset
Data (4 samples)	
Muon0_Run2023D_v1	/Muon0/Run2023D-22Sep2023_v1-v1/NANOAOD
Muon0_Run2023D_v2	/Muon0/Run2023D-22Sep2023_v2-v1/NANOAOD
Muon1_Run2023D_v1	/Muon1/Run2023D-22Sep2023_v1-v1/NANOAOD
Muon1_Run2023D_v2	/Muon1/Run2023D-22Sep2023_v2-v1/NANOAOD
Simulation (106 samples)	
DYto2Mu_MLL_105to160_amcatnloFXFX	/DYto2Mu-2Jets_MLL-105To160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
DYto2L_M_10to50_amcatnloFXFX	/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v4/NANOAODSIM /DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2_ext1-v4/NANOAODSIM
DYto2L_M_50_amcatnloFXFX	/DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v4/NANOAODSIM
DYto2L_M_50_0J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v4/NANOAODSIM
DYto2L_M_50_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v4/NANOAODSIM
DYto2L_M_50_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v5/NANOAODSIM
DYto2L_M_50_PTLL_40to100_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v1/NANOAODSIM
DYto2L_M_50_PTLL_40to100_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v1/NANOAODSIM
DYto2L_M_50_PTLL_100to200_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v1/NANOAODSIM
DYto2L_M_50_PTLL_100to200_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v1/NANOAODSIM
DYto2L_M_50_PTLL_200to400_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_200to400_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_400to600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v3/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v3/NANOAODSIM
DYto2L_M_50_PTLL_400to600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLL_600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM /DYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
GluGluHto2Tau_M125	/GluGluHToTauTau_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_Filtered	/GluGluHTo2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_UnFiltered	/GluGluHTo2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
GluGluHto2B_M125	/GluGluHto2B_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
VBFHto2B_M125	/VBFHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_UnFiltered	/VBFHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_Filtered	/VBFHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
ggZH_Hto2B_Zto2L	/ggZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
ggZH_Hto2B_Zto2Q	/ggZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WminusH_Hto2B_WtoLNu	/WminusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WminusH_Hto2B_Wto2Q	/WminusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_Filtered	/WminusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_UnFiltered	/WminusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WplusH_Hto2B_WtoLNu	/WplusH_Hto2B_WtoLNu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WplusH_Hto2B_Wto2Q	/WplusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_Filtered	/WplusHTo2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_UnFiltered	/WplusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TBbartoLplusNuBbar_s_channel_4FS	/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TbarBQ_t_channel_4FS	/TbarBQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TBbarQ_t_channel_4FS	/TBbarQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TbarWplustoLNu2Q	/TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TWminusto2L2Nu	/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM

Continued on next page

Table 6: NanoAOD 2023BPix datasets used in the analysis (continued)

Sample	NanoAOD dataset
TWminusto4Q	/TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
TWminustoLNu2Q	/TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
TT	/TT_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TTZH	/TTZH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TTZZ	/TTZZ_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
TTtoLNu2Q	/TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
VBFHto2Tau_M125	/VBFHToTauTau_M125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
WtoLNu_0J_amcatnloFXFX	/WtoLNu-2Jets_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WtoLNu_1J_amcatnloFXFX	/WtoLNu-2Jets_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WtoLNu_2J_amcatnloFXFX	/WtoLNu-2Jets_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WtoLNu_1J_madgraphMLM	/WtoLNu-4Jets_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WtoLNu_2J_madgraphMLM	/WtoLNu-4Jets_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WtoLNu_3J_madgraphMLM	/WtoLNu-4Jets_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WtoLNu_4J_madgraphMLM	/WtoLNu-4Jets_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WtoLNu_amcatnloFXFX	/WtoLNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
WtoLNu_madgraphMLM	/WtoLNu-4Jets_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WWtoLNu2Q_powheg	/WWtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WZto3LNu_powheg	/WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM /WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WZtoLNu2Q_powheg	/WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
ZZto2L2Nu_powheg	/ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
ZZto2Nu2Q_powheg	/ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WWW_4F	/WWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
WWZ_4F	/WWZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
WZZ	/WZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
ZZZ	/ZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
ZH_Hto2B_Zto2L	/ZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
ZH_Hto2B_Zto2Q	/ZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_Filtered	/ZHto2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_UnFiltered	/ZHto2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v5/NANOAODSIM
TTHtoNon2B_M125	/TTHtoNon2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
TTHto2B_M125	/TTHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
EWK_2L2J_madgraph_herwig	/EWK_2L2J_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK_2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_M2Mu-105to160_M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v3/NANOAODSIM
GluGluHto2Mu	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_MiNNLO	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powhegMiNNLO-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_tuneDown	/GluGluHto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_tuneUp	/GluGluHto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_amcatnlo	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_M120	/GluGluHto2Mu_M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM

Continued on next page

Table 6: NanoAOD 2023BPix datasets used in the analysis (continued)

Sample	**NanoAOD dataset**
GluGluHto2Mu_M130	/GluGluHto2Mu_M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_M125_powheg	/VBFHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_M120_amcatnlo	/VBFH_Hto2Mu_M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_M125_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_M130_amcatnlo	/VBFH_Hto2Mu_M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
ZH_Hto2Mu	/ZH_Hto2Mu_ZtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
WminusH_Hto2Mu	/WminusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
WplusH_Hto2Mu	/WplusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TTH_Hto2Mu	/TTH_Hto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM

Table 7: NanoAOD 2024 datasets used in the analysis

Sample	NanoAOD dataset
Data (96 samples)	
Muon0_Run2024C	/Muon0/Run2024C-MINIv6NANOv15-v1/NANOAOD
Muon0_Run2024D	/Muon0/Run2024D-MINIv6NANOv15-v1/NANOAOD
Muon0_Run2024E	/Muon0/Run2024E-MINIv6NANOv15-v1/NANOAOD
Muon0_Run2024F	/Muon0/Run2024F-MINIv6NANOv15-v1/NANOAOD
Muon0_Run2024G	/Muon0/Run2024G-MINIv6NANOv15-v1/NANOAOD
Muon0_Run2024H	/Muon0/Run2024H-MINIv6NANOv15-v1/NANOAOD
Muon0_Run2024I_v1	/Muon0/Run2024I-MINIv6NANOv15-v1/NANOAOD
Muon0_Run2024I_v2	/Muon0/Run2024I-MINIv6NANOv15_v2-v1/NANOAOD
Muon1_Run2024C	/Muon1/Run2024C-MINIv6NANOv15-v1/NANOAOD
Muon1_Run2024D	/Muon1/Run2024D-MINIv6NANOv15-v1/NANOAOD
Muon1_Run2024E	/Muon1/Run2024E-MINIv6NANOv15-v1/NANOAOD
Muon1_Run2024F	/Muon1/Run2024F-MINIv6NANOv15-v1/NANOAOD
Muon1_Run2024G	/Muon1/Run2024G-MINIv6NANOv15-v2/NANOAOD
Muon1_Run2024H	/Muon1/Run2024H-MINIv6NANOv15-v2/NANOAOD
Muon1_Run2024I_v1	/Muon1/Run2024I-MINIv6NANOv15-v1/NANOAOD
Muon1_Run2024I_v2	/Muon1/Run2024I-MINIv6NANOv15_v2-v1/NANOAOD
MuonEG_Run2024C	/MuonEG/Run2024C-MINIv6NANOv15-v1/NANOAOD
MuonEG_Run2024D	/MuonEG/Run2024D-MINIv6NANOv15-v1/NANOAOD
MuonEG_Run2024E	/MuonEG/Run2024E-MINIv6NANOv15-v1/NANOAOD
MuonEG_Run2024F	/MuonEG/Run2024F-MINIv6NANOv15-v1/NANOAOD
MuonEG_Run2024G	/MuonEG/Run2024G-MINIv6NANOv15-v2/NANOAOD
MuonEG_Run2024H	/MuonEG/Run2024H-MINIv6NANOv15-v2/NANOAOD
MuonEG_Run2024I_v1	/MuonEG/Run2024I-MINIv6NANOv15-v1/NANOAOD
MuonEG_Run2024I_v2	/MuonEG/Run2024I-MINIv6NANOv15_v2-v1/NANOAOD
ParkingVBF0_Run2024C	/ParkingVBF0/Run2024C-MINIv6NANOv15-v1/NANOAOD
ParkingVBF0_Run2024D	/ParkingVBF0/Run2024D-MINIv6NANOv15-v1/NANOAOD
ParkingVBF0/Run2024E	/ParkingVBF0/Run2024E-MINIv6NANOv15-v1/NANOAOD
ParkingVBF0/Run2024F	/ParkingVBF0/Run2024F-MINIv6NANOv15-v3/NANOAOD
ParkingVBF0/Run2024G	/ParkingVBF0/Run2024G-MINIv6NANOv15-v3/NANOAOD
ParkingVBF0/Run2024H	/ParkingVBF0/Run2024H-MINIv6NANOv15-v2/NANOAOD
ParkingVBF0_Run2024I_v1	/ParkingVBF0/Run2024I-MINIv6NANOv15-v2/NANOAOD
ParkingVBF0_Run2024I_v2	/ParkingVBF0/Run2024I-MINIv6NANOv15_v2-v2/NANOAOD
ParkingVBF1_Run2024C	/ParkingVBF1/Run2024C-MINIv6NANOv15-v1/NANOAOD
ParkingVBF1_Run2024D	/ParkingVBF1/Run2024D-MINIv6NANOv15-v1/NANOAOD
ParkingVBF1_Run2024E	/ParkingVBF1/Run2024E-MINIv6NANOv15-v1/NANOAOD
ParkingVBF1_Run2024F	/ParkingVBF1/Run2024F-MINIv6NANOv15-v3/NANOAOD
ParkingVBF1_Run2024G	/ParkingVBF1/Run2024G-MINIv6NANOv15-v4/NANOAOD
ParkingVBF1_Run2024H	/ParkingVBF1/Run2024H-MINIv6NANOv15-v3/NANOAOD
ParkingVBF1_Run2024I_v1	/ParkingVBF1/Run2024I-MINIv6NANOv15-v3/NANOAOD
ParkingVBF1_Run2024I_v2	/ParkingVBF1/Run2024I-MINIv6NANOv15_v2-v3/NANOAOD
ParkingVBF2_Run2024C	/ParkingVBF2/Run2024C-MINIv6NANOv15-v1/NANOAOD
ParkingVBF2_Run2024D	/ParkingVBF2/Run2024D-MINIv6NANOv15-v2/NANOAOD
ParkingVBF2_Run2024E	/ParkingVBF2/Run2024E-MINIv6NANOv15-v1/NANOAOD
ParkingVBF2_Run2024F	/ParkingVBF2/Run2024F-MINIv6NANOv15-v3/NANOAOD
ParkingVBF2_Run2024G	/ParkingVBF2/Run2024G-MINIv6NANOv15-v3/NANOAOD
ParkingVBF2_Run2024H	/ParkingVBF2/Run2024H-MINIv6NANOv15-v3/NANOAOD
ParkingVBF2_Run2024I_v1	/ParkingVBF2/Run2024I-MINIv6NANOv15-v3/NANOAOD
ParkingVBF2_Run2024I_v2	/ParkingVBF2/Run2024I-MINIv6NANOv15_v2-v2/NANOAOD
ParkingVBF3_Run2024C	/ParkingVBF3/Run2024C-MINIv6NANOv15-v1/NANOAOD
ParkingVBF3_Run2024D	/ParkingVBF3/Run2024D-MINIv6NANOv15-v2/NANOAOD
ParkingVBF3_Run2024E	/ParkingVBF3/Run2024E-MINIv6NANOv15-v1/NANOAOD
ParkingVBF3_Run2024F	/ParkingVBF3/Run2024F-MINIv6NANOv15-v3/NANOAOD
ParkingVBF3_Run2024G	/ParkingVBF3/Run2024G-MINIv6NANOv15-v3/NANOAOD

Continued on next page

Table 7: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
ParkingVBF3_Run2024H	/ParkingVBF3/Run2024H-MINIv6NANOv15-v3/NANOAOD
ParkingVBF3_Run2024I_v1	/ParkingVBF3/Run2024I-MINIv6NANOv15-v3/NANOAOD
ParkingVBF3_Run2024I_v2	/ParkingVBF3/Run2024I-MINIv6NANOv15_v2-v2/NANOAOD
ParkingVBF4_Run2024C	/ParkingVBF4/Run2024C-MINIv6NANOv15-v1/NANOAOD
ParkingVBF4_Run2024D	/ParkingVBF4/Run2024D-MINIv6NANOv15-v2/NANOAOD
ParkingVBF4_Run2024E	/ParkingVBF4/Run2024E-MINIv6NANOv15-v1/NANOAOD
ParkingVBF4_Run2024F	/ParkingVBF4/Run2024F-MINIv6NANOv15-v3/NANOAOD
ParkingVBF4_Run2024G	/ParkingVBF4/Run2024G-MINIv6NANOv15-v3/NANOAOD
ParkingVBF4_Run2024H	/ParkingVBF4/Run2024H-MINIv6NANOv15-v3/NANOAOD
ParkingVBF4_Run2024I_v1	/ParkingVBF4/Run2024I-MINIv6NANOv15-v3/NANOAOD
ParkingVBF4_Run2024I_v2	/ParkingVBF4/Run2024I-MINIv6NANOv15_v2-v2/NANOAOD
ParkingVBF5_Run2024C	/ParkingVBF5/Run2024C-MINIv6NANOv15-v1/NANOAOD
ParkingVBF5_Run2024D	/ParkingVBF5/Run2024D-MINIv6NANOv15-v2/NANOAOD
ParkingVBF5_Run2024E	/ParkingVBF5/Run2024E-MINIv6NANOv15-v1/NANOAOD
ParkingVBF5_Run2024F	/ParkingVBF5/Run2024F-MINIv6NANOv15-v3/NANOAOD
ParkingVBF5_Run2024G	/ParkingVBF5/Run2024G-MINIv6NANOv15-v3/NANOAOD
ParkingVBF5_Run2024H	/ParkingVBF5/Run2024H-MINIv6NANOv15-v3/NANOAOD
ParkingVBF5_Run2024I_v1	/ParkingVBF5/Run2024I-MINIv6NANOv15-v3/NANOAOD
ParkingVBF5_Run2024I_v2	/ParkingVBF5/Run2024I-MINIv6NANOv15_v2-v2/NANOAOD
ParkingVBF6_Run2024C	/ParkingVBF6/Run2024C-MINIv6NANOv15-v1/NANOAOD
ParkingVBF6_Run2024D	/ParkingVBF6/Run2024D-MINIv6NANOv15-v2/NANOAOD
ParkingVBF6_Run2024E	/ParkingVBF6/Run2024E-MINIv6NANOv15-v1/NANOAOD
ParkingVBF6_Run2024F	/ParkingVBF6/Run2024F-MINIv6NANOv15-v3/NANOAOD
ParkingVBF6_Run2024G	/ParkingVBF6/Run2024G-MINIv6NANOv15-v3/NANOAOD
ParkingVBF6_Run2024H	/ParkingVBF6/Run2024H-MINIv6NANOv15-v3/NANOAOD
ParkingVBF6_Run2024I_v1	/ParkingVBF6/Run2024I-MINIv6NANOv15-v3/NANOAOD
ParkingVBF6_Run2024I_v2	/ParkingVBF6/Run2024I-MINIv6NANOv15_v2-v2/NANOAOD
ParkingVBF7_Run2024C	/ParkingVBF7/Run2024C-MINIv6NANOv15-v1/NANOAOD
ParkingVBF7_Run2024D	/ParkingVBF7/Run2024D-MINIv6NANOv15-v2/NANOAOD
ParkingVBF7_Run2024E	/ParkingVBF7/Run2024E-MINIv6NANOv15-v1/NANOAOD
ParkingVBF7_Run2024F	/ParkingVBF7/Run2024F-MINIv6NANOv15-v3/NANOAOD
ParkingVBF7_Run2024G	/ParkingVBF7/Run2024G-MINIv6NANOv15-v3/NANOAOD
ParkingVBF7_Run2024H	/ParkingVBF7/Run2024H-MINIv6NANOv15-v3/NANOAOD
ParkingVBF7_Run2024I_v1	/ParkingVBF7/Run2024I-MINIv6NANOv15-v3/NANOAOD
ParkingVBF7_Run2024I_v2	/ParkingVBF7/Run2024I-MINIv6NANOv15_v2-v2/NANOAOD
Tau_Run2024C	/Tau/Run2024C-MINIv6NANOv15-v1/NANOAOD
Tau_Run2024D	/Tau/Run2024D-MINIv6NANOv15-v1/NANOAOD
Tau_Run2024E	/Tau/Run2024E-MINIv6NANOv15-v1/NANOAOD
Tau_Run2024F	/Tau/Run2024F-MINIv6NANOv15-v1/NANOAOD
Tau_Run2024G	/Tau/Run2024G-MINIv6NANOv15-v2/NANOAOD
Tau_Run2024H	/Tau/Run2024H-MINIv6NANOv15-v2/NANOAOD
Tau_Run2024I_v1	/Tau/Run2024I-MINIv6NANOv15-v1/NANOAOD
Tau_Run2024I_v2	/Tau/Run2024I-MINIv6NANOv15_v2-v1/NANOAOD
Simulation (177 samples)	
GluGluHto2Mu_M120	/GluGluH-Hto2Mu_Par-M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Mu	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
GluGluHto2Mu_amcatnlo	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
GluGluHto2Mu_MiNNLO	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Mu_tuneDown	/GluGluH-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
GluGluHto2Mu_tuneUp	/GluGluH-Hto2Mu_Par-M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Mu_M130	/GluGluH-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM

Continued on next page

Table 7: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
VBFHto2Mu_M125_powheg	/VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
VBFHto2Mu_M120	/VBF-Hto2Mu_Par-M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBF-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
VBFHto2Mu_M125_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBF-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
VBFHto2Mu_M130	/VBF-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBF-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
ggZH_Hto2Mu_ZtoAll_M125	/ggZH_HToMuMu_ZToAll_M125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Mu_m125_Flashsim	/VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER /VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2_ext1-v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
ZH_Hto2Mu	/ZH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WplusH_Hto2Mu	/WplusH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WminusH_Hto2Mu	/WminusH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
TTH_Hto2Mu	/TTH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /TTH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
DYto2Mu_M_50_0J_amcatnloFXFX	/DYto2Mu-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
DYto2Mu_M_50_1J_amcatnloFXFX	/DYto2Mu-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v1/NANOAODSIM
DYto2Mu_M_50_2J_amcatnloFXFX	/DYto2Mu-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
DYto2Mu_M_10to50_amcatnloFXFX	/DYto2Mu-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_M_50_amcatnloFXFX	/DYto2Mu-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v6/NANOAODSIM
DYto2Mu_MLL_105to160_amcatnloFXFX	/DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
DYto2Mu_MLL_105to160_amcatnloFXFX_Fil_VBF	/DYto2Mu-2Jets_Bin-MLL-105to160_Fil-VBF_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_M_50_0J_amcatnloFXFX	/DYto2E-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_M_50_1J_amcatnloFXFX	/DYto2E-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
DYto2E_M_50_2J_amcatnloFXFX	/DYto2E-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_M_10to50_amcatnloFXFX	/DYto2E-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_M_50_amcatnloFXFX	/DYto2E-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v4/NANOAODSIM
DYto2Tau_M_50_0J_amcatnloFXFX	/DYto2Tau-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v4/NANOAODSIM
DYto2Tau_M_50_1J_amcatnloFXFX	/DYto2Tau-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_M_50_2J_amcatnloFXFX	/DYto2Tau-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_M_10to50_amcatnloFXFX	/DYto2Tau-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_M_50_amcatnloFXFX	/DYto2Tau-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v7/NANOAODSIM
DYto2Mu_MLL_10to50_powheg	/DYto2Mu_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_50to120_powheg	/DYto2Mu_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_120to200_powheg	/DYto2Mu_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_200to400_powheg	/DYto2Mu_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_400to800_powheg	/DYto2Mu_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_800to1500_powheg	/DYto2Mu_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_1500to2500_powheg	/DYto2Mu_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_2500to4000_powheg	/DYto2Mu_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_4000to6000_powheg	/DYto2Mu_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_6000_powheg	/DYto2Mu_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_10to50_powheg	/DYto2Tau_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_50to120_powheg	/DYto2Tau_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_120to200_powheg	/DYto2Tau_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_200to400_powheg	/DYto2Tau_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_400to800_powheg	/DYto2Tau_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_800to1500_powheg	/DYto2Tau_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_1500to2500_powheg	/DYto2Tau_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_2500to4000_powheg	/DYto2Tau_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM

Continued on next page

Table 7: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
DYto2Tau_MLL_4000to6000_powheg	/DYto2Tau_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_6000_powheg	/DYto2Tau_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_10to50_powheg	/DYto2E_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_50to120_powheg	/DYto2E_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_120to200_powheg	/DYto2E_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_200to400_powheg	/DYto2E_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_400to800_powheg	/DYto2E_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_800to1500_powheg	/DYto2E_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_1500to2500_powheg	/DYto2E_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_2500to4000_powheg	/DYto2E_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_4000to6000_powheg	/DYto2E_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_6000_powheg	/DYto2E_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_50to130_powheg_minnlo	/DYto2Mu_Bin-MLL-50to130_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_130to200_powheg_minnlo	/DYto2Mu_Bin-MLL-130to200_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_200to400_powheg_minnlo	/DYto2Mu_Bin-MLL-200to400_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_400to600_powheg_minnlo	/DYto2Mu_Bin-MLL-400to600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_600to800_powheg_minnlo	/DYto2Mu_Bin-MLL-600to800_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_1000to1500_powheg_minnlo	/DYto2Mu_Bin-MLL-1000to1500_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_1500to2000_powheg_minnlo	/DYto2Mu_Bin-MLL-1500to2000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_2000to4000_powheg_minnlo	/DYto2Mu_Bin-MLL-2000to4000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_4000to6000_powheg_minnlo	/DYto2Mu_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_6000to13600_powheg_minnlo	/DYto2Mu_Bin-MLL-6000to13600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2L_M_50_amcatnloFXFX_Flashsim	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/fvaselli-flashsim_DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8-Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic-937767ad549695a10e40fd392f6c3633/USER
DYto2Mu_MLL_105to160_amcatnloFXFX_Flashsim	/DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/fvaselli-DY2Mu_2Jets_MLL-105to160-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-packedTrackGenJetAK4-limit-TrkGJ_eta-to-tracker-66034083b387d860d393385d18a4f59d/USER
DYto2Mu_M_50_amcatnloFXFX_Flashsim	/DYto2Mu-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/fvaselli-SoftActivity_NanoAOD_Test_v1-ace5d3f4ddab40d8fc0f18e7a8dbab44/USER
EWK_2L2J_madgraph_herwig	/EWK-2L2J_Bin-MLL-50-MJJ-120_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /EWK-2L2J_Bin-MLL-50-MJJ-120_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK-2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /EWK-2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
TTLNu-EWK_amcatnlo	/TTLNu-EWK_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WmWmJJ_EWK	/WmWmJJ-EWK_TuneCP5_13p6TeV-powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia_Flashsim	/EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/fvaselli-EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
ggZH_Hto2B_Zto2Q	/GluGluZH-Zto2Q-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ggZH_Hto2B_Zto2L	/GluGluZH-Zto2L-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2B_M125	/GluGluH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
GluGluHto2Tau_M125	/GluGluHto2Tau_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_Filtered	/GluGluH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_UnFiltered	/GluGluH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_Par-M-125_TuneCP5_13p6TeV_powheg-jhugen-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTHto2B_M125	/TTH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTHtoNon2B_M125	/TTH-HtoNon2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTZH_ZHto4B	/TTZH-ZHto4B_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2B_M125	/VBFH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBFH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
VBFHto2Tau_M125	/VBFH-HTo2Tau_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_UnFiltered	/VBFH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_Filtered	/VBFH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_Par-M-125_TuneCP5_13p6TeV_powheg-jhugen-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WminusH_Hto2B_WtoLNu	/WminusH-WtoLNu-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WplusH_Hto2B_WtoLNu	/WplusH-WtoLNu-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM

Continued on next page

Table 7: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
WminusHto2Tau_UncorrelatedDecay_Filtered	/WminusH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_UnFiltered	/WminusH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_Filtered	/WplusH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_UnFiltered	/WplusH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZH_Hto2B_Zto2L	/ZH-Zto2L-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZH_Hto2B_Zto2Q	/ZH-Zto2Q-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TBbartoLplusNuBbar_s_channel_4FS	/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarBQtoLNu_t_channel_4FS	/TbarBQtoLNu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarBQto2Q_t_channel_4FS	/TbarBQto2Q-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TBbarQtoLNu_t_channel_4FS	/TBbarQtoLNu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TBbarQto2Q_t_channel_4FS	/TBbarQto2Q-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarWplustoLNu2Q	/TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TWminusto2L2Nu	/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TWminusto4Q	/TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TWminustoLNu2Q	/TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTtoLNu2Q	/TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTWZ	/TTWZ_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTZZ_ZZto4B	/TTZZ-ZZto4B_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM/
TTZ_Zto2Q	/TTZ-ZtoQQ-1Jets_TuneCP5_13p6TeV_amcatnloFXFXold-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTto2L2Nu_Flashsim	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
TT_Flashsim	/TT_TuneCH3_13TeV-powheg-herwig7/fvaselli-flashsim_v9-f5e5c27141f6cbae6ce3c7d60feb1683/USER
WtoLNu_1J_madgraphMLM	/WtoLNu-4Jets_Bin-1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_2J_amcatnloFXFX	/WtoLNu-4Jets_Bin-2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_2J_madgraphMLM	/WtoLNu-4Jets_Bin-2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_3J_madgraphMLM	/WtoLNu-4Jets_Bin-3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_4J_madgraphMLM	/WtoLNu-4Jets_Bin-4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_40to100_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-40to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoLNu_PTLNu_40to100_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-40to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoLNu_PTLNu_100to200_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoLNu_PTLNu_100to200_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoLNu_PTLNu_200to400_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_200to400_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_400to600_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_400to600_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_600_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_600_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoENu_amcatnloFXFX	/WtoENu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoMuNu_amcatnloFXFX	/WtoMuNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoTauNu_amcatnloFXFX	/WtoTauNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWtoLNu2Q_powheg	/WWtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WZto3LNu_powheg	/WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM

Continued on next page

Table 7: NanoAOD 2024 datasets used in the analysis (continued)

Sample	**NanoAOD dataset**
WZtoLNu2Q_powheg	/WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZto2L2Nu_powheg	/ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZto2Nu2Q_powheg	/ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWW_4F	/WWW-4F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWZ_4F	/WWZ-4F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WZZ	/WZZ-5F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZZ	/ZZZ-5F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_Filtered	/ZH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v6/NANOAODSIM
Zto2Nu_HT_100to200	/Zto2Nu-4Jets_Bin-HT-100to200_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
Zto2Nu_HT_200to400	/Zto2Nu-4Jets_Bin-HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
Zto2Nu_HT_400to800	/Zto2Nu-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Nu_HT_800to1500	/Zto2Nu-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Nu_HT_1500to2500	/Zto2Nu-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Nu_HT_2500toInf	/Zto2Nu-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_100to400	/Zto2Q-4Jets_Bin-HT-100to400_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_400to800	/Zto2Q-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_800to1500	/Zto2Q-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_1500to2500	/Zto2Q-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_2500toInf	/Zto2Q-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM

Table 8: NanoAOD 2025 datasets used in the analysis

Sample	NanoAOD dataset
Data (96 samples)	
Muon0_Run2025B_v1	/Muon0/Run2025B-PromptReco-v1/NANOAOD
Muon0_Run2025C_v1	/Muon0/Run2025C-PromptReco-v1/NANOAOD
Muon0_Run2025C_v2	/Muon0/Run2025C-PromptReco-v2/NANOAOD
Muon0_Run2025D_v1	/Muon0/Run2025D-PromptReco-v1/NANOAOD
Muon0_Run2025E_v1	/Muon0/Run2025E-PromptReco-v1/NANOAOD
Muon0_Run2025F_v1	/Muon0/Run2025F-PromptReco-v1/NANOAOD
Muon0_Run2025F_v2	/Muon0/Run2025F-PromptReco-v2/NANOAOD
Muon0_Run2025G_v1	/Muon0/Run2025G-PromptReco-v1/NANOAOD
Muon1_Run2025B_v1	/Muon1/Run2025B-PromptReco-v1/NANOAOD
Muon1_Run2025C_v1	/Muon1/Run2025C-PromptReco-v1/NANOAOD
Muon1_Run2025C_v2	/Muon1/Run2025C-PromptReco-v2/NANOAOD
Muon1_Run2025D_v1	/Muon1/Run2025D-PromptReco-v1/NANOAOD
Muon1_Run2025E_v1	/Muon1/Run2025E-PromptReco-v1/NANOAOD
Muon1_Run2025F_v1	/Muon1/Run2025F-PromptReco-v1/NANOAOD
Muon1_Run2025F_v2	/Muon1/Run2025F-PromptReco-v2/NANOAOD
Muon1_Run2025G_v1	/Muon1/Run2025G-PromptReco-v1/NANOAOD
MuonEG_Run2025B_v1	/MuonEG/Run2025B-PromptReco-v1/NANOAOD
MuonEG_Run2025C_v1	/MuonEG/Run2025C-PromptReco-v1/NANOAOD
MuonEG_Run2025C_v2	/MuonEG/Run2025C-PromptReco-v2/NANOAOD
MuonEG_Run2025D_v1	/MuonEG/Run2025D-PromptReco-v1/NANOAOD
MuonEG_Run2025E_v1	/MuonEG/Run2025E-PromptReco-v1/NANOAOD
MuonEG_Run2025F_v1	/MuonEG/Run2025F-PromptReco-v1/NANOAOD
MuonEG_Run2025F_v2	/MuonEG/Run2025F-PromptReco-v2/NANOAOD
MuonEG_Run2025G_v1	/MuonEG/Run2025G-PromptReco-v1/NANOAOD
ParkingVBF0_Run2025B_v1	/ParkingVBF0/Run2025B-PromptReco-v1/NANOAOD
ParkingVBF0_Run2025C_v1	/ParkingVBF0/Run2025C-PromptReco-v1/NANOAOD
ParkingVBF0_Run2025C_v2	/ParkingVBF0/Run2025C-PromptReco-v2/NANOAOD
ParkingVBF0_Run2025D_v1	/ParkingVBF0/Run2025D-PromptReco-v1/NANOAOD
ParkingVBF0_Run2025E_v1	/ParkingVBF0/Run2025E-PromptReco-v1/NANOAOD
ParkingVBF0_Run2025F_v1	/ParkingVBF0/Run2025F-PromptReco-v1/NANOAOD
ParkingVBF0_Run2025F_v2	/ParkingVBF0/Run2025F-PromptReco-v2/NANOAOD
ParkingVBF0_Run2025G_v1	/ParkingVBF0/Run2025G-PromptReco-v1/NANOAOD
ParkingVBF1_Run2025B_v1	/ParkingVBF1/Run2025B-PromptReco-v1/NANOAOD
ParkingVBF1_Run2025C_v1	/ParkingVBF1/Run2025C-PromptReco-v1/NANOAOD
ParkingVBF1_Run2025C_v2	/ParkingVBF1/Run2025C-PromptReco-v2/NANOAOD
ParkingVBF1_Run2025D_v1	/ParkingVBF1/Run2025D-PromptReco-v1/NANOAOD
ParkingVBF1_Run2025E_v1	/ParkingVBF1/Run2025E-PromptReco-v1/NANOAOD
ParkingVBF1_Run2025F_v1	/ParkingVBF1/Run2025F-PromptReco-v1/NANOAOD
ParkingVBF1_Run2025F_v2	/ParkingVBF1/Run2025F-PromptReco-v2/NANOAOD
ParkingVBF1_Run2025G_v1	/ParkingVBF1/Run2025G-PromptReco-v1/NANOAOD
ParkingVBF2_Run2025B_v1	/ParkingVBF2/Run2025B-PromptReco-v1/NANOAOD
ParkingVBF2_Run2025C_v1	/ParkingVBF2/Run2025C-PromptReco-v1/NANOAOD
ParkingVBF2_Run2025C_v2	/ParkingVBF2/Run2025C-PromptReco-v2/NANOAOD
ParkingVBF2_Run2025D_v1	/ParkingVBF2/Run2025D-PromptReco-v1/NANOAOD
ParkingVBF2_Run2025E_v1	/ParkingVBF2/Run2025E-PromptReco-v1/NANOAOD
ParkingVBF2_Run2025F_v1	/ParkingVBF2/Run2025F-PromptReco-v1/NANOAOD
ParkingVBF2_Run2025F_v2	/ParkingVBF2/Run2025F-PromptReco-v2/NANOAOD
ParkingVBF2_Run2025G_v1	/ParkingVBF2/Run2025G-PromptReco-v1/NANOAOD
ParkingVBF3_Run2025B_v1	/ParkingVBF3/Run2025B-PromptReco-v1/NANOAOD
ParkingVBF3_Run2025C_v1	/ParkingVBF3/Run2025C-PromptReco-v1/NANOAOD
ParkingVBF3_Run2025C_v2	/ParkingVBF3/Run2025C-PromptReco-v2/NANOAOD
ParkingVBF3_Run2025D_v1	/ParkingVBF3/Run2025D-PromptReco-v1/NANOAOD
ParkingVBF3_Run2025E_v1	/ParkingVBF3/Run2025E-PromptReco-v1/NANOAOD

Continued on next page

Table 8: NanoAOD 2025 datasets used in the analysis (continued)

Sample	NanoAOD dataset
ParkingVBF3_Run2025F_v1	/ParkingVBF3/Run2025F-PromptReco-v1/NANOAOD
ParkingVBF3_Run2025F_v2	/ParkingVBF3/Run2025F-PromptReco-v2/NANOAOD
ParkingVBF3_Run2025G_v1	/ParkingVBF3/Run2025G-PromptReco-v1/NANOAOD
ParkingVBF4_Run2025B_v1	/ParkingVBF4/Run2025B-PromptReco-v1/NANOAOD
ParkingVBF4_Run2025C_v1	/ParkingVBF4/Run2025C-PromptReco-v1/NANOAOD
ParkingVBF4_Run2025C_v2	/ParkingVBF4/Run2025C-PromptReco-v2/NANOAOD
ParkingVBF4_Run2025D_v1	/ParkingVBF4/Run2025D-PromptReco-v1/NANOAOD
ParkingVBF4_Run2025E_v1	/ParkingVBF4/Run2025E-PromptReco-v1/NANOAOD
ParkingVBF4_Run2025F_v1	/ParkingVBF4/Run2025F-PromptReco-v1/NANOAOD
ParkingVBF4_Run2025F_v2	/ParkingVBF4/Run2025F-PromptReco-v2/NANOAOD
ParkingVBF4_Run2025G_v1	/ParkingVBF4/Run2025G-PromptReco-v1/NANOAOD
ParkingVBF5_Run2025B_v1	/ParkingVBF5/Run2025B-PromptReco-v1/NANOAOD
ParkingVBF5_Run2025C_v1	/ParkingVBF5/Run2025C-PromptReco-v1/NANOAOD
ParkingVBF5_Run2025C_v2	/ParkingVBF5/Run2025C-PromptReco-v2/NANOAOD
ParkingVBF5_Run2025D_v1	/ParkingVBF5/Run2025D-PromptReco-v1/NANOAOD
ParkingVBF5_Run2025E_v1	/ParkingVBF5/Run2025E-PromptReco-v1/NANOAOD
ParkingVBF5_Run2025F_v1	/ParkingVBF5/Run2025F-PromptReco-v1/NANOAOD
ParkingVBF5_Run2025F_v2	/ParkingVBF5/Run2025F-PromptReco-v2/NANOAOD
ParkingVBF5_Run2025G_v1	/ParkingVBF5/Run2025G-PromptReco-v1/NANOAOD
ParkingVBF6_Run2025B_v1	/ParkingVBF6/Run2025B-PromptReco-v1/NANOAOD
ParkingVBF6_Run2025C_v1	/ParkingVBF6/Run2025C-PromptReco-v1/NANOAOD
ParkingVBF6_Run2025C_v2	/ParkingVBF6/Run2025C-PromptReco-v2/NANOAOD
ParkingVBF6_Run2025D_v1	/ParkingVBF6/Run2025D-PromptReco-v1/NANOAOD
ParkingVBF6_Run2025E_v1	/ParkingVBF6/Run2025E-PromptReco-v1/NANOAOD
ParkingVBF6_Run2025F_v1	/ParkingVBF6/Run2025F-PromptReco-v1/NANOAOD
ParkingVBF6_Run2025F_v2	/ParkingVBF6/Run2025F-PromptReco-v2/NANOAOD
ParkingVBF6_Run2025G_v1	/ParkingVBF6/Run2025G-PromptReco-v1/NANOAOD
ParkingVBF7_Run2025B_v1	/ParkingVBF7/Run2025B-PromptReco-v1/NANOAOD
ParkingVBF7_Run2025C_v1	/ParkingVBF7/Run2025C-PromptReco-v1/NANOAOD
ParkingVBF7_Run2025C_v2	/ParkingVBF7/Run2025C-PromptReco-v2/NANOAOD
ParkingVBF7_Run2025D_v1	/ParkingVBF7/Run2025D-PromptReco-v1/NANOAOD
ParkingVBF7_Run2025E_v1	/ParkingVBF7/Run2025E-PromptReco-v1/NANOAOD
ParkingVBF7_Run2025F_v1	/ParkingVBF7/Run2025F-PromptReco-v1/NANOAOD
ParkingVBF7_Run2025F_v2	/ParkingVBF7/Run2025F-PromptReco-v2/NANOAOD
ParkingVBF7_Run2025G_v1	/ParkingVBF7/Run2025G-PromptReco-v1/NANOAOD
Tau_Run2025B_v1	/Tau/Run2025B-PromptReco-v1/NANOAOD
Tau_Run2025C_v1	/Tau/Run2025C-PromptReco-v1/NANOAOD
Tau_Run2025C_v2	/Tau/Run2025C-PromptReco-v2/NANOAOD
Tau_Run2025D_v1	/Tau/Run2025D-PromptReco-v1/NANOAOD
Tau_Run2025E_v1	/Tau/Run2025E-PromptReco-v1/NANOAOD
Tau_Run2025F_v1	/Tau/Run2025F-PromptReco-v1/NANOAOD
Tau_Run2025F_v2	/Tau/Run2025F-PromptReco-v2/NANOAOD
Tau_Run2025G_v1	/Tau/Run2025G-PromptReco-v1/NANOAOD
Simulation (177 samples)	
GluGluHto2Mu_M120	/GluGluH-Hto2Mu_Par-M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Mu	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
GluGluHto2Mu_amcatnlo	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
GluGluHto2Mu_MiNNLO	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Mu_tuneDown	/GluGluH-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
GluGluHto2Mu_tuneUp	/GluGluH-Hto2Mu_Par-M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Mu_M130	/GluGluH-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM

Continued on next page

Table 8: NanoAOD 2025 datasets used in the analysis (continued)

Sample	NanoAOD dataset
VBFHto2Mu_M125_powheg	/VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
VBFHto2Mu_M120	/VBF-Hto2Mu_Par-M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBF-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
VBFHto2Mu_M125_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBF-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
VBFHto2Mu_M130	/VBF-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBF-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
ggZH_Hto2Mu_ZtoAll_M125	/ggZH_HToMuMu_ZToAll_M125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Mu_m125_Flashsim	/VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER /VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2_ext1-v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
ZH_Hto2Mu	/ZH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WplusH_Hto2Mu	/WplusH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WminusH_Hto2Mu	/WminusH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
TTH_Hto2Mu	/TTH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /TTH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
DYto2Mu_M_50_0J_amcatnloFXFX	/DYto2Mu-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
DYto2Mu_M_50_1J_amcatnloFXFX	/DYto2Mu-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v1/NANOAODSIM
DYto2Mu_M_50_2J_amcatnloFXFX	/DYto2Mu-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
DYto2Mu_M_10to50_amcatnloFXFX	/DYto2Mu-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_M_50_amcatnloFXFX	/DYto2Mu-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v6/NANOAODSIM
DYto2Mu_MLL_105to160_amcatnloFXFX	/DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
DYto2Mu_MLL_105to160_amcatnloFXFX_Fil_VBF	/DYto2Mu-2Jets_Bin-MLL-105to160_Fil-VBF_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_M_50_0J_amcatnloFXFX	/DYto2E-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_M_50_1J_amcatnloFXFX	/DYto2E-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
DYto2E_M_50_2J_amcatnloFXFX	/DYto2E-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_M_10to50_amcatnloFXFX	/DYto2E-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_M_50_amcatnloFXFX	/DYto2E-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v4/NANOAODSIM
DYto2Tau_M_50_0J_amcatnloFXFX	/DYto2Tau-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v4/NANOAODSIM
DYto2Tau_M_50_1J_amcatnloFXFX	/DYto2Tau-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_M_50_2J_amcatnloFXFX	/DYto2Tau-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_M_10to50_amcatnloFXFX	/DYto2Tau-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_M_50_amcatnloFXFX	/DYto2Tau-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v7/NANOAODSIM
DYto2Mu_MLL_10to50_powheg	/DYto2Mu_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_50to120_powheg	/DYto2Mu_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_120to200_powheg	/DYto2Mu_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_200to400_powheg	/DYto2Mu_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_400to800_powheg	/DYto2Mu_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_800to1500_powheg	/DYto2Mu_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_1500to2500_powheg	/DYto2Mu_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_2500to4000_powheg	/DYto2Mu_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_4000to6000_powheg	/DYto2Mu_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_6000_powheg	/DYto2Mu_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_10to50_powheg	/DYto2Tau_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_50to120_powheg	/DYto2Tau_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_120to200_powheg	/DYto2Tau_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_200to400_powheg	/DYto2Tau_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_400to800_powheg	/DYto2Tau_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_800to1500_powheg	/DYto2Tau_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_1500to2500_powheg	/DYto2Tau_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_2500to4000_powheg	/DYto2Tau_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM

Continued on next page

Table 8: NanoAOD 2025 datasets used in the analysis (continued)

Sample	NanoAOD dataset
DYto2Tau_MLL_4000to6000_powheg	/DYto2Tau_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Tau_MLL_6000_powheg	/DYto2Tau_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_10to50_powheg	/DYto2E_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_50to120_powheg	/DYto2E_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_120to200_powheg	/DYto2E_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_200to400_powheg	/DYto2E_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_400to800_powheg	/DYto2E_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_800to1500_powheg	/DYto2E_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_1500to2500_powheg	/DYto2E_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_2500to4000_powheg	/DYto2E_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_4000to6000_powheg	/DYto2E_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2E_MLL_6000_powheg	/DYto2E_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_50to130_powheg_minnlo	/DYto2Mu_Bin-MLL-50to130_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_130to200_powheg_minnlo	/DYto2Mu_Bin-MLL-130to200_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_200to400_powheg_minnlo	/DYto2Mu_Bin-MLL-200to400_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_400to600_powheg_minnlo	/DYto2Mu_Bin-MLL-400to600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_600to800_powheg_minnlo	/DYto2Mu_Bin-MLL-600to800_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_1000to1500_powheg_minnlo	/DYto2Mu_Bin-MLL-1000to1500_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_1500to2000_powheg_minnlo	/DYto2Mu_Bin-MLL-1500to2000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_2000to4000_powheg_minnlo	/DYto2Mu_Bin-MLL-2000to4000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_4000to6000_powheg_minnlo	/DYto2Mu_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2Mu_MLL_6000to13600_powheg_minnlo	/DYto2Mu_Bin-MLL-6000to13600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
DYto2L_M_50_amcatnloFXFX_Flashsim	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/fvaselli-flashsim_DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8-Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic-937767ad549695a10e40fd392f6c3633/USER
DYto2Mu_MLL_105to160_amcatnloFXFX_Flashsim	/DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/fvaselli-DY2Mu_2Jets_MLL-105to160-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-packedTrackGenJetAK4-limit-TrkGJ_eta-to-tracker-66034083b387d860d393385d18a4f59d/USER
DYto2Mu_M_50_amcatnloFXFX_Flashsim	/DYto2Mu-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/fvaselli-SoftActivity_NanoAOD_Test_v1-ace5d3f4ddab40d8fc0f18e7a8dbab44/USER
EWK_2L2J_madgraph_herwig	/EWK-2L2J_Bin-MLL-50-MJJ-120_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /EWK-2L2J_Bin-MLL-50-MJJ-120_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK-2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /EWK-2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
TTLNu-EWK_amcatnlo	/TTLNu-EWK_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WmWmJJ_EWK	/WmWmJJ-EWK_TuneCP5_13p6TeV-powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia_Flashsim	/EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/fvaselli-EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
ggZH_Hto2B_Zto2Q	/GluGluZH-Zto2Q-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ggZH_Hto2B_Zto2L	/GluGluZH-Zto2L-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2B_M125	/GluGluH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /GluGluH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
GluGluHto2Tau_M125	/GluGluHto2Tau_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_Filtered	/GluGluH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Tau_UncorrelatedDecay_UnFiltered	/GluGluH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_Par-M-125_TuneCP5_13p6TeV_powheg-jhugen-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTHto2B_M125	/TTH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTHtoNon2B_M125	/TTH-HtoNon2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTZH_ZHto4B	/TTZH-ZHto4B_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2B_M125	/VBFH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM /VBFH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOAODSIM
VBFHto2Tau_M125	/VBFH-HTo2Tau_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_UnFiltered	/VBFH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Tau_UncorrelatedDecay_Filtered	/VBFH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_Par-M-125_TuneCP5_13p6TeV_powheg-jhugen-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WminusH_Hto2B_WtoLNu	/WminusH-WtoLNu-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WplusH_Hto2B_WtoLNu	/WplusH-WtoLNu-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM

Continued on next page

Table 8: NanoAOD 2025 datasets used in the analysis (continued)

Sample	NanoAOD dataset
WminusHto2Tau_UncorrelatedDecay_Filtered	/WminusH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WminusHto2Tau_UncorrelatedDecay_UnFiltered	/WminusH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_Filtered	/WplusH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WplusHto2Tau_UncorrelatedDecay_UnFiltered	/WplusH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZH_Hto2B_Zto2L	/ZH-Zto2L-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZH_Hto2B_Zto2Q	/ZH-Zto2Q-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TBbartoLplusNuBbar_s_channel_4FS	/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarBQtoLNu_t_channel_4FS	/TbarBQtoLNu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarBQto2Q_t_channel_4FS	/TbarBQto2Q-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TBbarQtoLNu_t_channel_4FS	/TBbarQtoLNu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TBbarQto2Q_t_channel_4FS	/TBbarQto2Q-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TbarWplustoLNu2Q	/TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TWminusto2L2Nu	/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TWminusto4Q	/TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TWminustoLNu2Q	/TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTtoLNu2Q	/TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTWZ	/TTWZ_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTZZ_ZZto4B	/TTZZ-ZZto4B_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM/
TTZ_Zto2Q	/TTZ-ZtoQQ-1Jets_TuneCP5_13p6TeV_amcatnloFXFXold-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
TTto2L2Nu_Flashsim	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
TT_Flashsim	/TT_TuneCH3_13TeV-powheg-herwig7/fvaselli-flashsim_v9-f5e5c27141f6cbae6ce3c7d60feb1683/USER
WtoLNu_1J_madgraphMLM	/WtoLNu-4Jets_Bin-1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_2J_amcatnloFXFX	/WtoLNu-4Jets_Bin-2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_2J_madgraphMLM	/WtoLNu-4Jets_Bin-2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_3J_madgraphMLM	/WtoLNu-4Jets_Bin-3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_4J_madgraphMLM	/WtoLNu-4Jets_Bin-4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_40to100_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-40to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoLNu_PTLNu_40to100_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-40to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoLNu_PTLNu_100to200_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoLNu_PTLNu_100to200_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoLNu_PTLNu_200to400_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_200to400_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_400to600_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_400to600_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_600_1J_amcatnloFXFX	/WtoLNu-2Jets_Bin-1J-PTLNu-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoLNu_PTLNu_600_2J_amcatnloFXFX	/WtoLNu-2Jets_Bin-2J-PTLNu-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WtoENu_amcatnloFXFX	/WtoENu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoMuNu_amcatnloFXFX	/WtoMuNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WtoTauNu_amcatnloFXFX	/WtoTauNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWtoLNu2Q_powheg	/WWtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WZto3LNu_powheg	/WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM

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Table 8: NanoAOD 2025 datasets used in the analysis (continued)

Sample	NanoAOD dataset
WZtoLNu2Q_powheg	/WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZto2L2Nu_powheg	/ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZto2Nu2Q_powheg	/ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWW_4F	/WWW-4F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WWZ_4F	/WWZ-4F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
WZZ	/WZZ-5F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZZZ	/ZZZ-5F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
ZHto2Tau_UncorrelatedDecay_Filtered	/ZH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v6/NANOAODSIM
Zto2Nu_HT_100to200	/Zto2Nu-4Jets_Bin-HT-100to200_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
Zto2Nu_HT_200to400	/Zto2Nu-4Jets_Bin-HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM
Zto2Nu_HT_400to800	/Zto2Nu-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Nu_HT_800to1500	/Zto2Nu-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Nu_HT_1500to2500	/Zto2Nu-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Nu_HT_2500toInf	/Zto2Nu-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_100to400	/Zto2Q-4Jets_Bin-HT-100to400_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_400to800	/Zto2Q-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_800to1500	/Zto2Q-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_1500to2500	/Zto2Q-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOAODSIM
Zto2Q_HT_2500toInf	/Zto2Q-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOAODSIM